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Align before Fuse: Vision and Language Representation Learning with Momentum Distillation

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Abstract

Large-scale vision and language representation learning has shown promising improvements on various vision-language tasks. Most existing methods employ a transformer-based multimodal encoder to jointly model visual tokens (region-based image features) and word tokens. Because the visual tokens and word tokens are unaligned, it is challenging for the multimodal encoder to learn image-text interactions. In this paper, we introduce a contrastive loss to ALign the image and text representations BEfore Fusing (ALBEF) them through cross-modal attention, which enables more grounded vision and language representation learning. Unlike most existing methods, our method does not require bounding box annotations nor high-resolution images. To improve learning from noisy web data, we propose momentum distillation, a self-training method which learns from pseudo-targets produced by a momentum model. We provide a theoretical analysis of ALBEF from a mutual information maximization perspective, showing that different training tasks can be interpreted as different ways to generate views for an image-text pair. ALBEF achieves state-of-the-art performance on multiple downstream vision-language tasks. On image-text retrieval, ALBEF outperforms methods that are pre-trained on orders of magnitude larger datasets. On VQA and NLVR^2, ALBEF achieves absolute improvements of 2.37% and 3.84% compared to the state-of-the-art, while enjoying faster inference speed. Code and models are available at https://github.com/salesforce/ALBEF.

1 Introduction

Vision-and-Language Pre-training (VLP) aims to learn multimodal representations from large-scale image-text pairs that can improve downstream Vision-and-Language (V+L) tasks. Most existing VLP methods (*e.g.* LXMERT [1], UNITER [2], OSCAR [3]) rely on pre-trained object detectors to extract region-based image features, and employ a multimodal encoder to fuse the image features with word tokens. The multimodal encoder is trained to solve tasks that require joint understanding of image and text, such as masked language modeling (MLM) and image-text matching (ITM).

While effective, this VLP framework suffers from several key limitations: (1) The image features and the word token embeddings reside in their own spaces, which makes it challenging for the multimodal encoder to learn to model their interactions; (2) The object detector is both annotation-expensive and compute-expensive, because it requires bounding box annotations during pre-training, and high-resolution (*e.g.* 600×1000) images during inference; (3) The widely used image-text datasets [4, 5] are collected from the web and are inherently noisy, and existing pre-training objectives such as MLM may overfit to the noisy text and degrade the model's generalization performance.

We propose ALign BEfore Fuse (ALBEF), a new VLP framework to address these limitations. We first encode the image and text independently with a detector-free image encoder and a text encoder. Then we use a multimodal encoder to fuse the image features with the text features through cross-modal attention. We introduce an intermediate image-text contrastive (ITC) loss on representations from the unimodal encoders, which serves three purposes: (1) it aligns the image features and the text features, making it easier for the multimodal encoder to perform cross-modal learning; (2) it

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融合前对齐：基于动量蒸馏的视觉与语言表征学习

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摘要

大规模视觉与语言表征学习已在各种视觉-语言任务上展现出令人期待的提升。现有大多数方法采用基于 Transformer 的多模态编码器来联合建模视觉标记（基于区域的图像特征）和词标记。由于视觉标记和词标记并未对齐，多模态编码器学习图像-文本交互具有挑战性。本文中，我们引入了一种对比损失，用于在通过跨模态注意力融合之前对齐图像和文本表示（ALBEF），从而实现更具依据的视觉与语言表征学习。不同于现有大多数方法，我们的方法既不需要边界框标注，也不需要高分辨率图像。为提升从噪声网络数据中的学习效果，我们提出了动量蒸馏，这是一种自训练方法，能够从动量模型生成的伪目标中学习。我们从互信息最大化的视角对 ALBEF 进行了理论分析，表明不同的训练任务可以被解释为为图像-文本对生成视图的不同方式。ALBEF 在多个下游视觉-语言任务上取得了最先进的性能。在图像-文本检索任务上，ALBEF 优于那些在大几个数量级的数据集上预训练的方法。在 VQA 和 NLVR^2 上，与当前最先进方法相比，ALBEF 分别实现了 2.37% 和 3.84% 的绝对提升，同时推理速度更快。代码和模型可在 https://github.com/salesforce/ALBEF 获取。

1 引言

视觉-语言预训练（VLP）旨在从大规模图像-文本对中学习多模态表示，从而提升下游视觉-语言（V+L）任务的表现。现有大多数 VLP 方法（*e.g.* LXMERT [1]、UNITER [2]、OSCAR [3]）依赖预训练的目标检测器来提取基于区域的图像特征，并采用多模态编码器将图像特征与词元进行融合。该多模态编码器通过解决需要对图像和文本进行联合理解的任务来训练，例如掩码语言建模（MLM）和图像-文本匹配（ITM）。

尽管有效，这种 VLP 框架仍存在几个关键局限：（1）图像特征和词元嵌入位于各自的空间中，这使得多模态编码器难以学习建模它们之间的交互；（2）目标检测器在标注和计算方面都代价高昂，因为它在预训练期间需要边界框标注，并且在推理期间需要高分辨率（*e.g.* 600×1000）图像；（3）广泛使用的图文数据集 [4, 5] 收集自网络，本质上存在噪声，而现有的预训练目标（如 MLM）可能会过拟合噪声文本，并降低模型的泛化性能。

我们提出了先对齐再融合（ALign BEfore Fuse，ALBEF），这是一种新的 VLP 框架，用于解决这些局限性。我们首先使用无检测器的图像编码器和文本编码器分别对图像和文本进行编码。然后，我们使用多模态编码器通过跨模态注意力将图像特征与文本特征进行融合。我们在来自单模态编码器的表示上引入了一个中间的图像-文本对比（ITC）损失，它有三个作用：（1）它对齐图像特征和文本特征，使多模态编码器更容易进行跨模态学习；（2）它

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improves the unimodal encoders to better understand the semantic meaning of images and texts; (3) it learns a common low-dimensional space to embed images and texts, which enables the image-text matching objective to find more informative samples through our contrastive hard negative mining.

To improve learning under noisy supervision, we propose Momentum Distillation (MoD), a simple method which enables the model to leverage a larger uncurated web dataset. During training, we keep a momentum version of the model by taking the moving-average of its parameters, and use the momentum model to generate pseudo-targets as additional supervision. With MoD, the model is not penalized for producing other reasonable outputs that are different from the web annotation. We show that MoD not only improves pre-training, but also downstream tasks with clean annotations.

We provide theoretical justifications on ALBEF from the perspective of mutual information maximization. Specifically, we show that ITC and MLM maximize a lower bound on the mutual information between different views of an image-text pair, where the views are generated by taking partial information from each pair. From this perspective, our momentum distillation can be interpreted as generating new views with semantically similar samples. Therefore, ALBEF learns vision-language representations that are invariant to semantic-preserving transformations.

We demonstrate the effectiveness of ALBEF on various downstream V+L tasks including image-text retrieval, visual question answering, visual reasoning, visual entailment, and weakly-supervised visual grounding. ALBEF achieves substantial improvements over existing state-of-the-art methods. On image-text retrieval, it outperforms methods that are pre-trained on orders of magnitude larger datasets (CLIP [6] and ALIGN [7]). On VQA and NLVR^2, it achieves absolute improvements of 2.37% and 3.84% compared to the state-of-the-art method VILLA [8], while enjoying much faster inference speed. We also provide quantitative and qualitative analysis on ALBEF using Grad-CAM [9], which reveals its ability to perform accurate object, attribute and relationship grounding implicitly.

2 Related Work

2.1 Vision-Language Representation Learning

Most existing work on vision-language representation learning fall into two categories. The first category focuses on modelling the interactions between image and text features with transformer-based multimodal encoders [10, 11, 12, 13, 1, 14, 15, 2, 3, 16, 8, 17, 18]. Methods in this category achieve superior performance on downstream V+L tasks that require complex reasoning over image and text (*e.g.* NLVR^2 [19], VQA [20]), but most of them require high-resolution input images and pre-trained object detectors. A recent method [21] improves inference speed by removing the object detector, but results in lower performance. The second category focuses on learning separate unimodal encoders for image and text [22, 23, 6, 7]. The recent CLIP [6] and ALIGN [7] perform pre-training on massive noisy web data using a contrastive loss, one of the most effective loss for representation learning [24, 25, 26, 27]. They achieve remarkable performance on image-text retrieval tasks, but lack the ability to model more complex interactions between image and text for other V+L tasks [21].

ALBEF unifies the two categories, leading to strong unimodal and multimodal representations with superior performance on both retrieval and reasoning tasks. Furthermore, ALBEF does not require object detectors, a major computation bottleneck for many existing methods [1, 2, 3, 8, 17].

2.2 Knowledge Distillation

Knowledge distillation [28] aims to improve a student model's performance by distilling knowledge from a teacher model, usually through matching the student's prediction with the teacher's. While most methods focus on distilling knowledge from a pre-trained teacher model [28, 29, 30, 31, 32], online distillation [33, 34] simultaneously trains multiple models and use their ensemble as the teacher. Our momentum distillation can be interpreted as a form of online self-distillation, where a temporal ensemble of the student model is used as the teacher. Similar ideas have been explored in semi-supervised learning [35], label noise learning [36], and very recently in contrastive learning [37]. Different from existing studies, we theoretically and experimentally show that momentum distillation is a generic learning algorithm that can improve the model's performance on many V+L tasks.

3 ALBEF Pre-training

In this section, we first introduce the model architecture (Section 3.1). Then we delineate the pre-training objectives (Section 3.2), followed by the proposed momentum distillation (Section 3.3). Lastly we describe the pre-training datasets (Section 3.4) and implementation details (Section 3.5).

改进单模态编码器，使其更好地理解图像和文本的语义含义；（3）学习一个共同的低维空间来嵌入图像和文本，从而使图像-文本匹配目标能够通过我们的对比式困难负样本挖掘找到更具信息量的样本。

为提升噪声监督下的学习效果，我们提出了动量蒸馏（MoD），这是一种简单方法，使模型能够利用更大规模的未经筛选的网络数据集。在训练过程中，我们通过对模型参数取移动平均来保留一个动量版本的模型，并使用该动量模型生成伪目标作为额外监督。借助 MoD，当模型产生与网络标注不同但同样合理的其他输出时，不会受到惩罚。我们表明，MoD 不仅能改善预训练效果，还能提升带有干净标注的下游任务表现。

我们从互信息最大化的角度为 ALBEF 提供理论依据。具体而言，我们表明，ITC 和 MLM 最大化了图像-文本对不同视图之间互信息的下界，其中这些视图是通过从每个图像-文本对中提取部分信息生成的。从这一角度来看，我们的动量蒸馏可以被解释为利用语义相似的样本生成新的视图。因此，ALBEF 学习到的视觉-语言表示对保持语义的变换具有不变性。

我们在各种下游 V+L 任务上展示了 ALBEF 的有效性，包括图文检索、视觉问答、视觉推理、视觉蕴含以及弱监督视觉定位。ALBEF 相较于现有最先进方法取得了显著提升。在图文检索任务上，它优于那些在大几个数量级的数据集上进行预训练的方法（CLIP [6] 和 ALIGN [7]）。在 VQA 和 NLVR^2 上，与最先进方法 VILLA [8] 相比，它分别取得了 2.37% 和 3.84% 的绝对提升，同时推理速度快得多。我们还使用 Grad-CAM [9] 对 ALBEF 进行了定量和定性分析，揭示了其能够隐式地进行准确的目标、属性和关系定位。

2 相关工作

2.1 视觉-语言表征学习

现有关于视觉-语言表征学习的大多数工作可分为两类。第一类侧重于使用基于 Transformer 的多模态编码器对图像和文本特征之间的交互进行建模 [10, 11, 12, 13, 1, 14, 15, 2, 3, 16, 8, 17, 18]。这一类方法在需要对图像和文本进行复杂推理的下游 V+L 任务上取得了优越性能（*e.g.* NLVR^2 [19]、VQA [20]），但其中大多数需要高分辨率输入图像和预训练的目标检测器。最近的一种方法 [21] 通过移除目标检测器提高了推理速度，但导致性能下降。第二类侧重于为图像和文本学习分离的单模态编码器 [22, 23, 6, 7]。近期的 CLIP [6] 和 ALIGN [7] 使用对比损失在大规模噪声网络数据上进行预训练，对比损失是表征学习中最有效的损失之一 [24, 25, 26, 27]。它们在图文检索任务上取得了显著性能，但缺乏为其他 V+L 任务建模图像和文本之间更复杂交互的能力 [21]。

ALBEF 统一了这两类方法，从而获得了强大的单模态和多模态表示，并在检索和推理任务上都取得了优越的性能。此外，ALBEF 不需要目标检测器，而目标检测器是许多现有方法的主要计算瓶颈 [1, 2, 3, 8, 17]。

2.2 知识蒸馏

知识蒸馏 [28] 旨在通过从教师模型中蒸馏知识来提升学生模型的性能，通常是通过使学生的预测与教师的预测相匹配来实现。尽管大多数方法侧重于从预训练的教师模型中蒸馏知识 [28, 29, 30, 31, 32]，在线蒸馏 [33, 34] 则同时训练多个模型，并使用它们的集成作为教师。我们的动量蒸馏可以被解释为一种在线自蒸馏形式，其中学生模型的时间集成被用作教师。类似的思想已在半监督学习 [35]、标签噪声学习 [36] 以及最近的对比学习 [37] 中得到探索。不同于现有研究，我们从理论和实验上表明，动量蒸馏是一种通用学习算法，能够提升模型在许多 V+L 任务上的性能。

3 ALBEF 预训练

在本节中，我们首先介绍模型架构（第 3.1 节）。然后阐述预训练目标（第 3.2 节），接着介绍所提出的动量蒸馏（第 3.3 节）。最后，我们描述预训练数据集（第 3.4 节）和实现细节（第 3.5 节）。

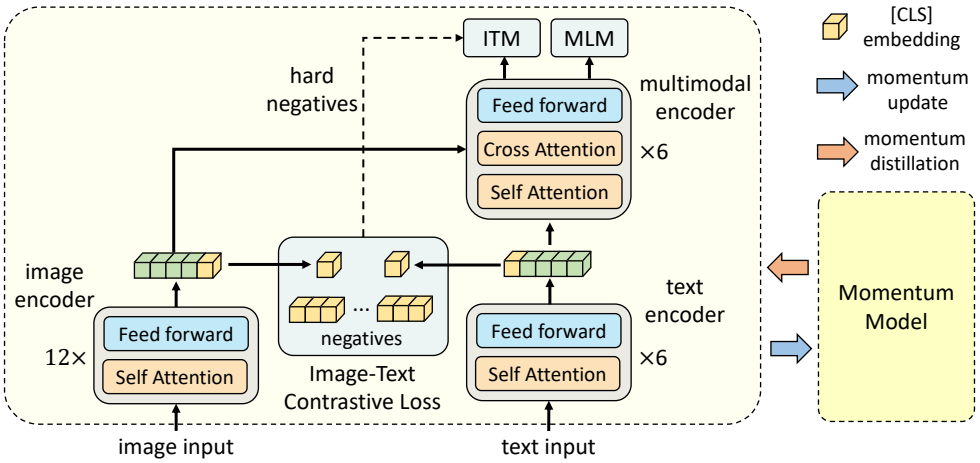


Figure 1: **Illustration of ALBEF.** It consists of an image encoder, a text encoder, and a multimodal encoder. We propose an image-text contrastive loss to align the unimodal representations of an image-text pair before fusion. An image-text matching loss (using in-batch hard negatives mined through contrastive similarity) and a masked-language-modeling loss are applied to learn multimodal interactions between image and text. In order to improve learning with noisy data, we generate pseudo-targets using the momentum model (a moving-average version of the base model) as additional supervision during training.

3.1 Model Architecture

As illustrated in Figure 1, ALBEF contains an image encoder, a text encoder, and a multimodal encoder. We use a 12-layer visual transformer ViT-B/16 [38] as the image encoder, and initialize it with weights pre-trained on ImageNet-1k from [31]. An input image I is encoded into a sequence of embeddings: $\{\boldsymbol{v}_{\text{cls}}, \boldsymbol{v}_1, ..., \boldsymbol{v}_N\}$, where v_{cls} is the embedding of the `[CLS]` token. We use a 6-layer transformer [39] for both the text encoder and the multimodal encoder. The text encoder is initialized using the first 6 layers of the $\text{BERT}_{\text{base}}$ [40] model, and the multimodal encoder is initialized using the last 6 layers of the $\text{BERT}_{\text{base}}$. The text encoder transforms an input text T into a sequence of embeddings $\{\boldsymbol{w}_{\text{cls}}, \boldsymbol{w}_1, ..., \boldsymbol{w}_N\}$, which is fed to the multimodal encoder. The image features are fused with the text features through cross attention at each layer of the multimodal encoder.

3.2 Pre-training Objectives

We pre-train ALBEF with three objectives: image-text contrastive learning (ITC) on the unimodal encoders, masked language modeling (MLM) and image-text matching (ITM) on the multimodal encoder. We improve ITM with online contrastive hard negative mining.

Image-Text Contrastive Learning aims to learn better unimodal representations before fusion. It learns a similarity function $s = g_v(\boldsymbol{v}_{\text{cls}})^\top g_w(\boldsymbol{w}_{\text{cls}})$, such that parallel image-text pairs have higher similarity scores. g_v and g_w are linear transformations that map the `[CLS]` embeddings to normalized lower-dimensional (256-d) representations. Inspired by MoCo [24], we maintain two queues to store the most recent M image-text representations from the momentum unimodal encoders. The normalized features from the momentum encoders are denoted as $g'_v(\boldsymbol{v}'_{\text{cls}})$ and $g'_w(\boldsymbol{w}'_{\text{cls}})$. We define $s(I,T) = g_v(\boldsymbol{v}_{\text{cls}})^\top g'_w(\boldsymbol{w}'_{\text{cls}})$ and $s(T,I) = g_w(\boldsymbol{w}_{\text{cls}})^\top g'_v(\boldsymbol{v}'_{\text{cls}})$.

For each image and text, we calculate the softmax-normalized image-to-text and text-to-image similarity as:

$$p_m^{\text{i2t}}(I) = \frac{\exp(s(I,T_m)/\tau)}{\sum_{m=1}^{M}\exp(s(I,T_m)/\tau)}, \quad p_m^{\text{t2i}}(T) = \frac{\exp(s(T,I_m)/\tau)}{\sum_{m=1}^{M}\exp(s(T,I_m)/\tau)} \tag{1}$$

where τ is a learnable temperature parameter. Let $\boldsymbol{y}^{\text{i2t}}(I)$ and $\boldsymbol{y}^{\text{t2i}}(T)$ denote the ground-truth one-hot similarity, where negative pairs have a probability of 0 and the positive pair has a probability of 1. The image-text contrastive loss is defined as the cross-entropy H between $\boldsymbol{p}$ and $\boldsymbol{y}$:

$$\mathcal{L}_{\text{itc}} = \frac{1}{2}\mathbb{E}_{(I,T)\sim D}\left[\text{H}(\boldsymbol{y}^{\text{i2t}}(I), \boldsymbol{p}^{\text{i2t}}(I)) + \text{H}(\boldsymbol{y}^{\text{t2i}}(T), \boldsymbol{p}^{\text{t2i}}(T))\right] \tag{2}$$

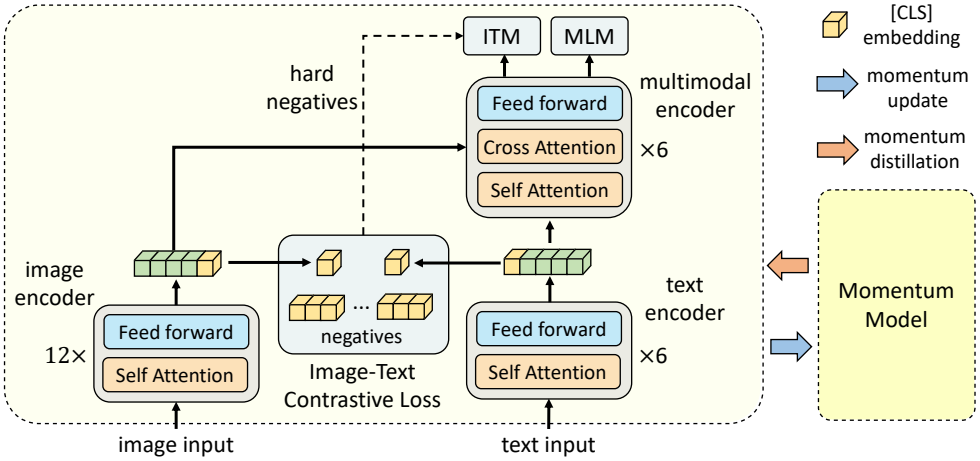


图1：ALBEF 示意图。它由图像编码器、文本编码器和多模态编码器组成。我们提出了一种图像-文本对比损失，用于在融合之前对齐图像-文本对的单模态表示。通过对比相似度挖掘的批内困难负样本，应用图像-文本匹配损失，以及掩码语言建模损失，以学习图像与文本之间的多模态交互。为了提升对噪声数据的学习效果，我们使用动量模型（基础模型的移动平均版本）生成伪目标，作为训练过程中的额外监督。

3.1 模型架构

如图 1 所示，ALBEF 包含一个图像编码器、一个文本编码器和一个多模态编码器。我们使用 12 层视觉 Transformer ViT-B/16 [38] 作为图像编码器，并采用来自 [31] 的在 ImageNet-1k 上预训练的权重对其进行初始化。输入图像 I 被编码为一系列嵌入：$\{\boldsymbol{v}_{\text{cls}}, \boldsymbol{v}_1, ..., \boldsymbol{v}_N\}$，其中 v_{cls} 是 [CLS] token 的嵌入。我们为文本编码器和多模态编码器均使用 6 层 Transformer [39]。文本编码器使用 $\text{BERT}_{\text{base}}$ [40] 模型的前 6 层进行初始化，多模态编码器使用 $\text{BERT}_{\text{base}}$ 的后 6 层进行初始化。文本编码器将输入文本 T 转换为一系列嵌入 $\{\boldsymbol{w}_{\text{cls}}, \boldsymbol{w}_1, ..., \boldsymbol{w}_N\}$，并将其输入到多模态编码器中。在多模态编码器的每一层中，图像特征通过交叉注意力与文本特征进行融合。

3.2 预训练目标

我们使用三个目标对 ALBEF 进行预训练：在单模态编码器上进行图像-文本对比学习（ITC），在多模态编码器上进行掩码语言建模（MLM）和图像-文本匹配（ITM）。我们通过在线对比式困难负样本挖掘来改进 ITM。

图文对比学习旨在融合之前学习更好的单模态表示。它学习一个相似度函数 $s = g_v(\boldsymbol{v}_{\text{cls}})^\top g_w(\boldsymbol{w}_{\text{cls}})$，使得匹配的图文对具有更高的相似度分数。$g_v$ 和 g_w 是线性变换，用于将 [CLS] 嵌入映射到归一化的低维（256 维）表示。受 MoCo [24] 启发，我们维护两个队列，用于存储来自动量单模态编码器的最近 M 个图文表示。来自动量编码器的归一化特征表示为 $g'_v(\boldsymbol{v}'_{\text{cls}})$ 和 $g'_w(\boldsymbol{w}'_{\text{cls}})$。我们定义 $s(I,T) = g_v(\boldsymbol{v}_{\text{cls}})^\top g'_w(\boldsymbol{w}'_{\text{cls}})$ 和 $s(T,I) = g_w(\boldsymbol{w}_{\text{cls}})^\top g'_v(\boldsymbol{v}'_{\text{cls}})$。

对于每张图像和文本，我们计算经过 softmax 归一化的图像到文本和文本到图像相似度如下：

$$p_m^{\text{i2t}}(I) = \frac{\exp(s(I,T_m)/\tau)}{\sum_{m=1}^{M}\exp(s(I,T_m)/\tau)}, \quad p_m^{\text{t2i}}(T) = \frac{\exp(s(T,I_m)/\tau)}{\sum_{m=1}^{M}\exp(s(T,I_m)/\tau)} \tag{1}$$

其中 τ 是一个可学习的温度参数。令 $\boldsymbol{y}^{\text{i2t}}(I)$ 和 $\boldsymbol{y}^{\text{t2i}}(T)$ 表示真实的独热相似度，其中负样本对的概率为 0，正样本对的概率为 1。图文对比损失定义为 $\boldsymbol{p}$ 和 $\boldsymbol{y}$ 之间的交叉熵 H：

$$\mathcal{L}_{\text{itc}} = \frac{1}{2}\mathbb{E}_{(I,T)\sim D}\left[\text{H}(\boldsymbol{y}^{\text{i2t}}(I), \boldsymbol{p}^{\text{i2t}}(I)) + \text{H}(\boldsymbol{y}^{\text{t2i}}(T), \boldsymbol{p}^{\text{t2i}}(T))\right] \tag{2}$$

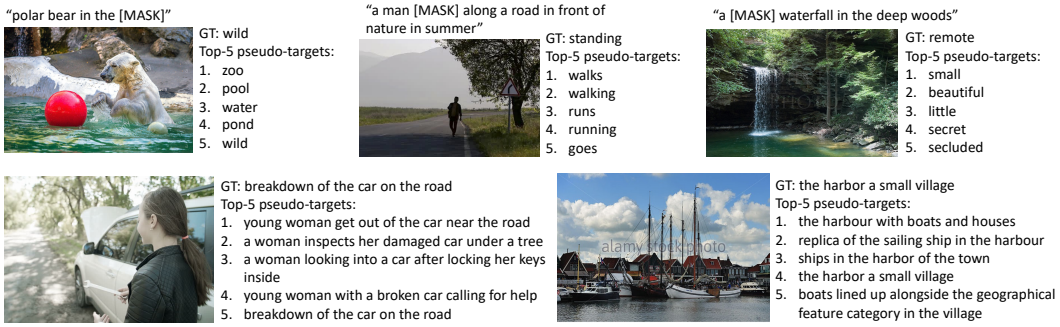


Figure 2: Examples of the pseudo-targets for MLM (1st row) and ITC (2nd row). The pseudo-targets can capture visual concepts that are not described by the ground-truth text (*e.g.* "beautiful waterfall", "young woman").

Masked Language Modeling utilizes both the image and the contextual text to predict the masked words. We randomly mask out the input tokens with a probability of 15% and replace them with the special token `[MASK]`[1]. Let $\hat{T}$ denote a masked text, and $\boldsymbol{p}^{\text{msk}}(I, \hat{T})$ denote the model's predicted probability for a masked token. MLM minimizes a cross-entropy loss:

$$\mathcal{L}_{\text{mlm}} = \mathbb{E}_{(I,\hat{T})\sim D} \text{H}(\boldsymbol{y}^{\text{msk}}, \boldsymbol{p}^{\text{msk}}(I, \hat{T})) \tag{3}$$

where $\boldsymbol{y}^{\text{msk}}$ is a one-hot vocabulary distribution where the ground-truth token has a probability of 1.

Image-Text Matching predicts whether a pair of image and text is positive (matched) or negative (not matched). We use the multimodal encoder's output embedding of the `[CLS]` token as the joint representation of the image-text pair, and append a fully-connected (FC) layer followed by softmax to predict a two-class probability p^{itm}. The ITM loss is:

$$\mathcal{L}_{\text{itm}} = \mathbb{E}_{(I,T)\sim D} \text{H}(\boldsymbol{y}^{\text{itm}}, \boldsymbol{p}^{\text{itm}}(I, T)) \tag{4}$$

where $\boldsymbol{y}^{\text{itm}}$ is a 2-dimensional one-hot vector representing the ground-truth label.

We propose a strategy to sample hard negatives for the ITM task with zero computational overhead. A negative image-text pair is hard if they share similar semantics but differ in fine-grained details. We use the contrastive similarity from Equation 1 to find in-batch hard negatives. For each image in a mini-batch, we sample one negative text from the same batch following the contrastive similarity distribution, where texts that are more similar to the image have a higher chance to be sampled. Likewise, we also sample one hard negative image for each text.

The full pre-training objective of ALBEF is:

$$\mathcal{L} = \mathcal{L}_{\text{itc}} + \mathcal{L}_{\text{mlm}} + \mathcal{L}_{\text{itm}} \tag{5}$$

3.3 Momentum Distillation

The image-text pairs used for pre-training are mostly collected from the web and they tend to be noisy. Positive pairs are usually weakly-correlated: the text may contain words that are unrelated to the image, or the image may contain entities that are not described in the text. For ITC learning, negative texts for an image may also match the image's content. For MLM, there may exist other words different from the annotation that describes the image equally well (or better). However, the one-hot labels for ITC and MLM penalize all negative predictions regardless of their correctness.

To address this, we propose to learn from pseudo-targets generated by the momentum model. The momentum model is a continuously-evolving teacher which consists of exponential-moving-average versions of the unimodal and multimodal encoders. During training, we train the base model such that its predictions match the ones from the momentum model. Specifically, for ITC, we first compute the image-text similarity using features from the momentum unimodal encoders as $s'(I,T) = g'_v(\boldsymbol{v}'_{\text{cls}})^\top g'_w(\boldsymbol{w}'_{\text{cls}})$ and $s'(T,I) = g'_w(\boldsymbol{w}_{\text{cls}})^\top g'_v(\boldsymbol{v}'_{\text{cls}})$. Then we compute soft pseudo-targets $\boldsymbol{q}^{\text{i2t}}$ and $\boldsymbol{q}^{\text{t2i}}$ by replacing s with s' in Equation 1. The ITC_{MoD} loss is defined as:

$$\mathcal{L}^{\text{mod}}_{\text{itc}} = (1-\alpha)\mathcal{L}_{\text{itc}} + \frac{\alpha}{2}\mathbb{E}_{(I,T)\sim D}\left[\text{KL}(\boldsymbol{q}^{\text{i2t}}(I) \parallel \boldsymbol{p}^{\text{i2t}}(I)) + \text{KL}(\boldsymbol{q}^{\text{t2i}}(T) \parallel \boldsymbol{p}^{\text{t2i}}(T))\right] \tag{6}$$

[1]following BERT, the replacements are 10% random tokens, 10% unchanged, and 80% `[MASK]`

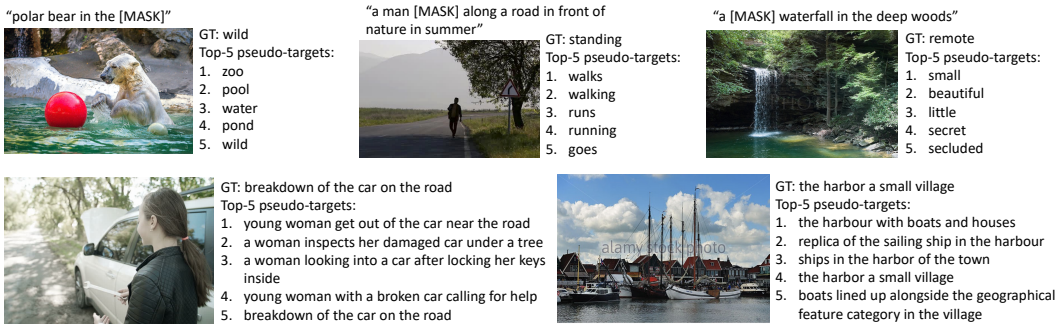


图 2：MLM（第 1 行）和 ITC（第 2 行）的伪目标示例。伪目标可以捕捉真实文本未描述的视觉概念（*e.g.*“美丽的瀑布”、“年轻女性”）。

掩码语言建模利用图像和上下文文本来预测被遮蔽的词。我们以 15% 的概率随机遮蔽输入标记，并将其替换为特殊标记 [MASK][1]。令 $\hat{T}$ 表示被遮蔽的文本，$\boldsymbol{p}^{\text{msk}}(I, \hat{T})$ 表示模型对被遮蔽标记的预测概率。MLM 最小化交叉熵损失：

$$\mathcal{L}_{\text{mlm}} = \mathbb{E}_{(I,\hat{T})\sim D} \text{H}(\boldsymbol{y}^{\text{msk}}, \boldsymbol{p}^{\text{msk}}(I, \hat{T})) \tag{3}$$

其中 $\boldsymbol{y}^{\text{msk}}$ 是一个独热词汇分布，其中真实标记的概率为 1。

图文匹配用于预测一对图像和文本是正样本（匹配）还是负样本（不匹配）。我们使用多模态编码器输出的 [CLS] 标记嵌入作为图文对的联合表示，并接入一个全连接（FC）层，随后通过 softmax 预测二分类概率 p^{itm}。ITM 损失为：

$$\mathcal{L}_{\text{itm}} = \mathbb{E}_{(I,T)\sim D} \text{H}(\boldsymbol{y}^{\text{itm}}, \boldsymbol{p}^{\text{itm}}(I, T)) \tag{4}$$

其中 $\boldsymbol{y}^{\text{itm}}$ 是表示真实标签 l 的二维 one-hot 向量 阿贝尔。

我们提出了一种用于 ITM 任务的困难负样本采样策略，且不会带来任何计算开销。如果一个负图像-文本对共享相似的语义但在细粒度细节上存在差异，则它就是困难的。我们使用公式 1 中的对比相似度来寻找批内困难负样本。对于一个小批量中的每张图像，我们按照对比相似度分布从同一批量中采样一个负文本，其中与图像越相似的文本被采样的概率越高。同样地，我们也为每个文本采样一张困难负图像。

ALBEF 的完整预训练目标为：

$$\mathcal{L} = \mathcal{L}_{\text{itc}} + \mathcal{L}_{\text{mlm}} + \mathcal{L}_{\text{itm}} \tag{5}$$

3.3 动量蒸馏

用于预训练的图文对大多从网络收集而来，往往含有噪声。正样本对通常是弱相关的：文本可能包含与图像无关的词，或者图像可能包含文本中未描述的实体。对于 ITC 学习而言，一张图像的负样本文本也可能与该图像内容匹配。对于 MLM 而言，可能存在与标注不同但同样能很好（甚至更好）描述图像的其他词。然而，ITC 和 MLM 的独热标签会惩罚所有负向预测，而不管它们是否正确。

为了解决这一问题，我们提出从由动量模型生成的伪目标中学习。动量模型是一个持续演化的教师模型，由单模态编码器和多模态编码器的指数移动平均版本组成。在训练过程中，我们训练基础模型，使其预测与动量模型的预测相匹配。具体而言，对于 ITC，我们首先使用动量单模态编码器的特征计算图像-文本相似度，记为 $s'(I,T) = g'_v(\boldsymbol{v}'_{\text{cls}})^\top g'_w(\boldsymbol{w}'_{\text{cls}})$ 和 $s'(T,I) = g'_w(\boldsymbol{w}_{\text{cls}})^\top g'_v(\boldsymbol{v}'_{\text{cls}})$。然后，我们通过在公式 1 中用 s' 替换 s 来计算软伪目标 $\boldsymbol{q}^{\text{i2t}}$ 和 $\boldsymbol{q}^{\text{t2i}}$。$\text{ITC}_{\text{MoD}}$ 损失定义为：

$$\mathcal{L}_{\text{itc}}^{\text{mod}} = (1-\alpha)\mathcal{L}_{\text{itc}} + \frac{\alpha}{2}\mathbb{E}_{(I,T)\sim D}\left[\text{KL}(\boldsymbol{q}^{\text{i2t}}(I) \parallel \boldsymbol{p}^{\text{i2t}}(I)) + \text{KL}(\boldsymbol{q}^{\text{t2i}}(T) \parallel \boldsymbol{p}^{\text{t2i}}(T))\right] \tag{6}$$

1遵循 BERT，替换项为 10% 随机词元、10% 保持不变、80% [MASK]

Similarly, for MLM, let $\boldsymbol{q}^{\text{msk}}(I, \hat{T})$ denote the momentum model's prediction probability for the masked token, the MLM_{MoD} loss is:

$$\mathcal{L}_{\text{mlm}}^{\text{mod}} = (1-\alpha)\mathcal{L}_{\text{mlm}} + \alpha \mathbb{E}_{(I,\hat{T})\sim D} \text{KL}(\boldsymbol{q}^{\text{msk}}(I, \hat{T}) \parallel \boldsymbol{p}^{\text{msk}}(I, \hat{T})) \tag{7}$$

In Figure 2, we show examples of the top-5 candidates from the pseudo-targets, which effectively capture relevant words/texts for an image. More examples can be found in Appendix.

We also apply MoD to the downstream tasks. The final loss for each task is a weighted combination of the original task's loss and the KL-divergence between the model's prediction and the pseudo-targets. For simplicity, we set the weight $\alpha = 0.4$ for all pre-training and downstream tasks [2].

3.4 Pre-training Datasets

Following UNITER [2], we construct our pre-training data using two web datasets (Conceptual Captions [4], SBU Captions [5]) and two in-domain datasets (COCO [41] and Visual Genome [42]). The total number of unique images is 4.0M, and the number of image-text pairs is 5.1M. To show that our method is scalable with larger-scale web data, we also include the much noisier Conceptual 12M dataset [43], increasing the total number of images to 14.1M [3]. Details are in Appendix.

3.5 Implementation Details

Our model consists of a $\text{BERT}_{\text{base}}$ with 123.7M parameters and a ViT-B/16 with 85.8M parameters. We pre-train the model for 30 epochs using a batch size of 512 on 8 NVIDIA A100 GPUs. We use the AdamW [44] optimizer with a weight decay of 0.02. The learning rate is warmed-up to $1e^{-4}$ in the first 1000 iterations, and decayed to $1e^{-5}$ following a cosine schedule. During pre-training, we take random image crops of resolution 256×256 as input, and also apply RandAugment[4] [45]. During fine-tuning, we increase the image resolution to 384×384 and interpolate the positional encoding of image patches following [38]. The momentum parameter for updating the momentum model is set as 0.995, and the size of the queue used for image-text contrastive learning is set as 65,536. We linearly ramp-up the distillation weight α from 0 to 0.4 within the 1st epoch.

4 A Mutual Information Maximization Perspective

In this section, we provide an alternative perspective of ALBEF and show that it maximizes a lower bound on the mutual information (MI) between different "views" of an image-text pair. ITC, MLM, and MoD can be interpreted as different ways to generate the views.

Formally, we define two random variables a and b as two different views of a data point. In self-supervised learning [24, 25, 46], a and b are two augmentations of the same image. In vision-language representation learning, we consider a and b as different variations of an image-text pair that capture its semantic meaning. We aim to learn representations invariant to the change of view. This can be achieved by maximizing the MI between a and b. In practice, we maximize a lower bound on $\text{MI}(a, b)$ by minimizing the InfoNCE loss [47] defined as:

$$\mathcal{L}_{\text{NCE}} = -\mathbb{E}_{p(a,b)} \left[\log \frac{\exp(s(a,b))}{\sum_{\hat{b}\in\hat{B}} \exp(s(a,\hat{b}))} \right] \tag{8}$$

where $s(a, b)$ is a scoring function (*e.g.*, a dot product between two representations), and $\hat{B}$ contains the positive sample b and $|\hat{B}| - 1$ negative samples drawn from a proposal distribution.

Our ITC loss with one-hot labels (Equation 2) can be re-written as:

$$\mathcal{L}_{\text{itc}} = -\frac{1}{2}\mathbb{E}_{p(I,T)} \big[\log \frac{\exp(s(I,T)/\tau)}{\sum_{m=1}^{M} \exp(s(I,T_m)/\tau)} + \log \frac{\exp(s(T,I)/\tau)}{\sum_{m=1}^{M} \exp(s(T,I_m)/\tau)} \big] \tag{9}$$

Minimizing $\mathcal{L}_{\text{itc}}$ can be seen as maximizing a symmetric version of InfoNCE. Hence, ITC considers the two individual modalities (*i.e.*, I and T) as the two views of an image-text pair, and trains the unimodal encoders to maximize the MI between the image and text views for the positive pairs.

[2] our experiments show that $\alpha = 0.3, 0.4, 0.5$ yield similar performance, with $\alpha = 0.4$ slightly better

[3] some urls provided by the web datasets have become invalid

[4] we remove color changes from RandAugment because the text often contains color information

类似地，对于 MLM，令 $\boldsymbol{q}^{\text{msk}}(I,\hat{T})$ 表示动量模型对被掩蔽标记的预测概率，MLM_{MoD} 损失为：

$$\mathcal{L}_{\text{mlm}}^{\text{mod}} = (1-\alpha)\mathcal{L}_{\text{mlm}} + \alpha\mathbb{E}_{(I,\hat{T})\sim D}\text{KL}(\boldsymbol{q}^{\text{msk}}(I,\hat{T}) \parallel \boldsymbol{p}^{\text{msk}}(I,\hat{T})) \tag{7}$$

在图2中，我们展示了伪目标中排名前5的候选示例，它们能够有效捕捉与图像相关的词语/文本。更多示例见附录。

我们还将 MoD 应用于下游任务。每个任务的最终损失是原始任务损失与模型预测和伪目标之间 KL 散度的加权组合。为简单起见，我们将所有预训练和下游任务的权重 $\alpha = 0.4$ [2]。

3.4 预训练数据集

按照 UNITER [2]，我们使用两个网络数据集（Conceptual Captions [4]、SBU Captions [5]）和两个领域内数据集（COCO [41] 和 Visual Genome [42]）来构建我们的预训练数据。唯一图像的总数为 4.0M，图像-文本对的数量为 5.1M。为了表明我们的方法能够扩展到更大规模的网络数据，我们还纳入了噪声更多的 Conceptual 12M 数据集 [43]，使图像总数增加到 14.1M [3]。详情见附录。

3.5 实现细节

我们的模型由一个具有 123.7M 参数的 $\text{BERT}_{\text{base}}$ 和一个具有 85.8M 参数的 ViT-B/16 组成。我们在 8 块 NVIDIA A100 GPU 上使用 512 的批量大小对模型进行 30 个 epoch 的预训练。我们使用 AdamW [44] 优化器，权重衰减为 0.02。学习率在前 1000 次迭代中预热至 $1e^{-4}$，随后按照余弦调度衰减至 $1e^{-5}$。在预训练期间，我们将分辨率为 256 × 256 的随机图像裁剪作为输入，并应用 RandAugment[4] [45]。在微调期间，我们将图像分辨率提高到 384 × 384，并按照 [38] 对图像块的位置编码进行插值。用于更新动量模型的动量参数设为 0.995，用于图文对比学习的队列大小设为 65,536。我们在第 1 个 epoch 内将蒸馏权重 α 从 0 线性提升到 0.4。

4 互信息最大化视角

在本节中，我们提供了 ALBEF 的另一种视角，并表明它最大化了图文对不同“视图”之间互信息（MI）的下界。ITC、MLM 和 MoD 可以被解释为生成这些视图的不同方式。

形式上，我们将两个随机变量 a 和 b 定义为一个数据点的两个不同视图。在自监督学习 [24, 25, 46] 中，a 和 b 是同一图像的两种增强。在视觉-语言表征学习中，我们将 a 和 b 视为图像-文本对的不同变体，用于捕捉其语义含义。我们的目标是学习对视图变化保持不变的表征。这可以通过最大化 a 和 b 之间的互信息来实现。在实践中，我们通过最小化 [47] 中定义的 InfoNCE 损失来最大化 MI(a, b) 的下界：

$$\mathcal{L}_{\text{NCE}} = -\mathbb{E}_{p(a,b)}\left[\log \frac{\exp(s(a,b))}{\sum_{\hat{b}\in\hat{B}}\exp(s(a,\hat{b}))}\right] \tag{8}$$

其中，$s(a,b)$ 是一个评分函数（*e.g.*，即两个表示之间的点积），$\hat{B}$ 包含正样本 b 以及从提议分布中抽取的 $|\hat{B}|-1$ 个负样本。

采用 one-hot 标签的 ITC 损失（公式 2）可以重写为：

$$\mathcal{L}_{\text{itc}} = -\frac{1}{2}\mathbb{E}_{p(I,T)}\big[\log \frac{\exp(s(I,T)/\tau)}{\sum_{m=1}^{M}\exp(s(I,T_m)/\tau)} + \log \frac{\exp(s(T,I)/\tau)}{\sum_{m=1}^{M}\exp(s(T,I_m)/\tau)}\big] \tag{9}$$

最小化 $\mathcal{L}_{\text{itc}}$ 可被视为最大化 InfoNCE 的对称版本。因此，ITC 将两种单独模态（*i.e.*、I 和 T）视为图像-文本对的两个视图，并训练单模态编码器以最大化正样本对中图像视图与文本视图之间的 MI。

[2] our experiments show that $\alpha = 0.3, 0.4, 0.5$ yield similar performance, with $\alpha = 0.4$ slightly better

[3] some urls provided by the web datasets have become invalid

[4] we remove color changes from RandAugment because the text often contains color information

As shown in [48], we can also interpret MLM as maximizing the MI between a masked word token and its masked context (*i.e.* image + masked text). Specifically, we can re-write the MLM loss with one-hot labels (Equation 3) as

$$\mathcal{L}_{\text{mlm}} = -\mathbb{E}_{p(I,\hat{T})}\big[\log \frac{\exp(\psi(y^{\text{msk}})^\top f(I,\hat{T}))}{\sum_{y\in\mathcal{V}}\exp(\psi(y)^\top f(I,\hat{T}))}\big] \tag{10}$$

where $\psi(y) : \mathcal{V} \to \mathbb{R}^d$ is a lookup function in the multimodal encoder's output layer that maps a word token y into a vector and $\mathcal{V}$ is the full vocabulary set, and $f(I,\hat{T})$ is a function that returns the final hidden state of the multimodal encoder corresponding to the masked context. Hence, MLM considers the two views of an image-text pair to be: (1) a randomly selected word token, and (2) the image + the contextual text with that word masked.

Both ITC and MLM generate views by taking partial information from an image-text pair, through either modality separation or word masking. Our momentum distillation can be considered as generating alternative views from the entire proposal distribution. Take ITC_{MoD} in Equation 6 as an example, minimizing $\text{KL}(\boldsymbol{p}^{\text{i2t}}(I), \boldsymbol{q}^{\text{i2t}}(I))$ is equivalent to minimizing the following objective:

$$-\sum_m q_m^{\text{i2t}}(I)\log p_m^{\text{i2t}}(I) = -\sum_m \frac{\exp(s'(I,T_m)/\tau)}{\sum_{m=1}^M \exp(s'(I,T_m)/\tau)} \log \frac{\exp(s(I,T_m)/\tau)}{\sum_{m=1}^M \exp(s(I,T_m)/\tau)} \tag{11}$$

It maximizes $\text{MI}(I,T_m)$ for texts that share similar semantic meaning with the image I because those texts would have larger $q_m^{\text{i2t}}(I)$. Similarly, ITC_{MoD} also maximizes $\text{MI}(I_m,T)$ for images that are similar to T. We can follow the same method to show that MLM_{MoD} generates alternative views $y' \in \mathcal{V}$ for the masked word y^{msk}, and maximizes the MI between y' and $(I,\hat{T})$. Therefore, our momentum distillation can be considered as performing data augmentation to the original views. The momentum model generates a diverse set of views that are absent in the original image-text pairs, and encourages the base model to learn representations that capture view-invariant semantic information.

5 Downstream V+L Tasks

We adapt the pre-trained model to five downstream V+L tasks. We introduce each task and our fine-tuning strategy below. Details of the datasets and fine-tuning hyperparameters are in Appendix.

Image-Text Retrieval contains two subtasks: image-to-text retrieval (TR) and text-to-image retrieval (IR). We evaluate ALBEF on the Flickr30K [49] and COCO benchmarks, and fine-tune the pre-trained model using the training samples from each dataset. For zero-shot retrieval on Flickr30K, we evaluate with the model fine-tuned on COCO. During fine-tuning, we jointly optimize the ITC loss (Equation 2) and the ITM loss (Equation 4). ITC learns an image-text scoring function based on similarity of unimodal features, whereas ITM models the fine-grained interaction between image and text to predict a matching score. Since the downstream datasets contain multiple texts for each image, we change the ground-truth label of ITC to consider multiple positives in the queue, where each positive has a ground-truth probability of $1/\#\text{positives}$. During inference, we first compute the feature similarity score s_{itc} for all image-text pairs. Then we take the top-k candidates and calculate their ITM score s_{itm} for ranking. Because k can be set to be very small, our inference speed is much faster than methods that require computing the ITM score for all image-text pairs [2, 3, 8].

Visual Entailment (SNLI-VE[5] [51]) is a fine-grained visual reasoning task to predict whether the relationship between an image and a text is entailment, neutral, or contradictory. We follow UNITER [2] and consider VE as a three-way classification problem, and predict the class probabilities using a multi-layer perceptron (MLP) on the multimodal encoder's representation of the `[CLS]` token.

Visual Question Answering (VQA [52]) requires the model to predict an answer given an image and a question. Different from existing methods that formulate VQA as a multi-answer classification problem [53, 2], we consider VQA as an answer generation problem, similar to [54]. Specifically, we use a 6-layer transformer decoder to generate the answer. As shown in Figure 3a, the auto-regressive answer decoder receives the multimodal embeddings through cross attention, and a start-of-sequence token (`[CLS]`) is used as the decoder's initial input token. Likewise, an end-of-sequence token (`[SEP]`) is appended to the end of decoder outputs which indicates the completion of generation.

[5] results on SNLI-VE should be interpreted with caution because its test data has been reported to be noisy [50]

如 [48] 所示，我们也可以将 MLM 解释为最大化被掩码词元与其掩码上下文（*i.e.* 图像 + 掩码文本）之间的 MI。具体而言，我们可以将带有 one-hot 标签的 MLM 损失（公式 3）重写为

$$\mathcal{L}_{\mathrm{mlm}} = -\mathbb{E}_{p(I,\hat{T})}\big[\log \frac{\exp(\psi(y^{\mathrm{msk}})^\top f(I,\hat{T}))}{\sum_{y\in\mathcal{V}}\exp(\psi(y)^\top f(I,\hat{T}))}\big] \tag{10}$$

其中，$\psi(y):\mathcal{V}\to\mathbb{R}^d$ 是多模态编码器输出层中的查找函数，用于将词元 y 映射为向量，$\mathcal{V}$ 是完整词汇表集合；$f(I,\hat{T})$ 是一个函数，用于返回与被掩码上下文对应的多模态编码器的最终隐藏状态。因此，MLM 将图像-文本对的两个视图视为：（1）随机选择的词元，以及（2）图像 + 与该词被掩码的上下文文本。

ITC 和 MLM 都通过从图像-文本对中获取部分信息来生成视图，方式可以是模态分离或词遮蔽。我们的动量蒸馏可以被视为从整个提议分布中生成替代视图。以公式 6 中的 $\mathrm{ITC}_{\mathrm{MoD}}$ 为例，最小化 $\mathrm{KL}(\boldsymbol{p}^{\mathrm{i2t}}(I),\boldsymbol{q}^{\mathrm{i2t}}(I))$ 等价于最小化以下目标：

$$-\sum_m q_m^{\mathrm{i2t}}(I)\log p_m^{\mathrm{i2t}}(I) = -\sum_m \frac{\exp(s'(I,T_m)/\tau)}{\sum_{m=1}^M \exp(s'(I,T_m)/\tau)}\log\frac{\exp(s(I,T_m)/\tau)}{\sum_{m=1}^M\exp(s(I,T_m)/\tau)} \tag{11}$$

它最大化与图像 I 具有相似语义的文本的 $\mathrm{MI}(I,T_m)$，因为这些文本会具有更大的 $q_m^{\mathrm{i2t}}(I)$。类似地，$\mathrm{ITC}_{\mathrm{MoD}}$ 也会最大化与 T 相似的图像的 $\mathrm{MI}(I_m,T)$。我们可以采用相同的方法来证明，$\mathrm{MLM}_{\mathrm{MoD}}$ 会为被遮蔽词 y^{msk} 生成替代视图 $y'\in\mathcal{V}$，并最大化 y' 与 $(I,\hat{T})$ 之间的 MI。因此，我们的动量蒸馏可以被视为对原始视图进行数据增强。动量模型生成了一组原始图文对中不存在的多样化视图，并鼓励基础模型学习能够捕捉视图不变语义信息的表征。

5 个下游 V+L 任务

我们将预训练模型适配到五个下游 V+L 任务。下面介绍各项任务以及我们的微调策略。数据集和微调超参数的详细信息见附录。

图文检索包含两个子任务：以图检文（image-to-text retrieval, TR）和以文检图（text-to-image retrieval, IR）。我们在 Flickr30K [49] 和 COCO 基准上评估 ALBEF，并使用各数据集的训练样本对预训练模型进行微调。对于 Flickr30K 上的零样本检索，我们使用在 COCO 上微调后的模型进行评估。在微调过程中，我们联合优化 ITC 损失（公式 2）和 ITM 损失（公式 4）。ITC 基于单模态特征的相似性学习图文评分函数，而 ITM 则建模图像与文本之间的细粒度交互，以预测匹配分数。由于下游数据集中每张图像包含多条文本，我们将 ITC 的真实标签改为考虑队列中的多个正样本，其中每个正样本的真实概率为 1/#positives。在推理过程中，我们首先计算所有图文对的特征相似度分数 s_{itc}。然后取 top-k 个候选项，并计算它们的 ITM 分数 s_{itm} 用于排序。由于 k 可以设置得非常小，我们的推理速度远快于那些需要为所有图文对计算 ITM 分数的方法 [2, 3, 8]。

视觉蕴含（SNLI-VE[5] [51]）是一项细粒度的视觉推理任务，用于预测图像与文本之间的关系是蕴含、中性还是矛盾。我们遵循 UNITER [2]，将 VE 视为一个三分类问题，并使用多层感知机（MLP）基于多模态编码器对 [CLS] 标记的表示来预测类别概率。

视觉问答（VQA [52]）要求模型在给定一张图像和一个问题的情况下预测答案。不同于将 VQA 表述为多答案分类问题的现有方法 [53, 2]，我们将 VQA 视为一个答案生成问题，类似于 [54]。具体而言，我们使用一个 6 层 Transformer 解码器来生成答案。如图 3a 所示，自回归答案解码器通过交叉注意力接收多模态嵌入，并使用序列起始标记（[CLS]）作为解码器的初始输入标记。同样，序列结束标记（[SEP]）被附加到解码器输出的末尾，用于表示生成完成。

[5] results on SNLI-VE should be interpreted with caution because its test data has been reported to be noisy [50]

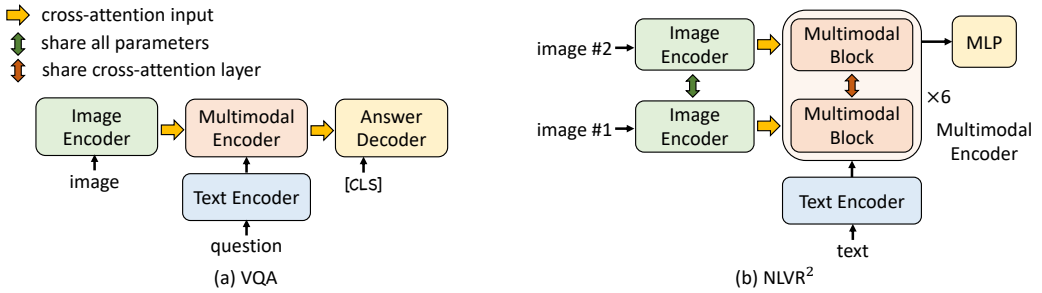


Figure 3: The model architecture for VQA and NLVR^2. For VQA, we append an auto-regressive decoder to generate the answer given the image-question embeddings. For NLVR^2, we replicate the transformer block within each layer of multimodal encoder to enable reasoning over two images.

The answer decoder is initialized using the pre-trained weights from the multimodal encoder, and finetuned with a conditional language-modeling loss. For a fair comparison with existing methods, we constrain the decoder to only generate from the 3,192 candidate answers [55] during inference.

Natural Language for Visual Reasoning (NLVR^2 [19]) requires the model to predict whether a text describes a pair of images. We extend our multimodal encoder to enable reasoning over two images. As shown in Figure 3b, each layer of the multimodal encoder is replicated to have two consecutive transformer blocks, where each block contains a self-attention layer, a cross-attention layer, and a feed-forward layer (see Figure 1). The two blocks within each layer are initialized using the same pre-trained weights, and the two cross-attention layers share the same linear projection weights for the keys and values. During training, the two blocks receive two sets of image embeddings for the image pair. We append a MLP classifier on the multimodal encoder's `[CLS]` representation for prediction.

For NLVR^2, we perform an additional pre-training step to prepare the new multimodal encoder for encoding an image-pair. We design a text-assignment (TA) task as follows: given a pair of images and a text, the model needs to assign the text to either the first image, the second image, or none of them. We consider it as a three-way classification problem, and use a FC layer on the `[CLS]` representation to predict the assignment. We pre-train with TA for only 1 epoch using the 4M images (Section 3.4).

Visual Grounding aims to localize the region in an image that corresponds to a specific textual description. We study the weakly-supervised setting, where no bounding box annotations are available. We perform experiments on the RefCOCO+ [56] dataset, and fine-tune the model using only image-text supervision following the same strategy as image-text retrieval. During inference, we extend Grad-CAM [9] to acquire heatmaps, and use them to rank the detected proposals provided by [53].

6 Experiments

6.1 Evaluation on the Proposed Methods

First, we evaluate the effectiveness of the proposed methods (*i.e.* image-text contrastive learning, contrastive hard negative mining, and momentum distillation). Table 1 shows the performance of the downstream tasks with different variants of our method. Compared to the baseline pre-training tasks (MLM+ITM), adding ITC substantially improves the pre-trained model's performance across

#Pre-train Images	Training tasks	TR (flickr test)	IR (flickr test)	SNLI-VE (test)	NLVR^2 (test-P)	VQA (test-dev)
4M	MLM + ITM	93.96	88.55	77.06	77.51	71.40
	ITC + MLM + ITM	96.55	91.69	79.15	79.88	73.29
	ITC + MLM + ITM_{hard}	97.01	92.16	79.77	80.35	73.81
	ITC_{MoD} + MLM + ITM_{hard}	97.33	92.43	79.99	80.34	74.06
	Full (ITC_{MoD} + MLM_{MoD} + ITM_{hard})	97.47	92.58	80.12	80.44	74.42
	ALBEF (Full + $\text{MoD}_{\text{Downstream}}$)	97.83	92.65	80.30	80.50	74.54
14M	ALBEF	98.70	94.07	80.91	83.14	75.84

Table 1: Evaluation of the proposed methods on four downstream V+L tasks. For text-retrieval (TR) and image-retrieval (IR), we report the average of R@1, R@5 and R@10. ITC: image-text contrastive learning. MLM: masked language modeling. ITM_{hard}: image-text matching with contrastive hard negative mining. MoD: momentum distillation. $\text{MoD}_{\text{Downstream}}$: momentum distillation on downstream tasks.

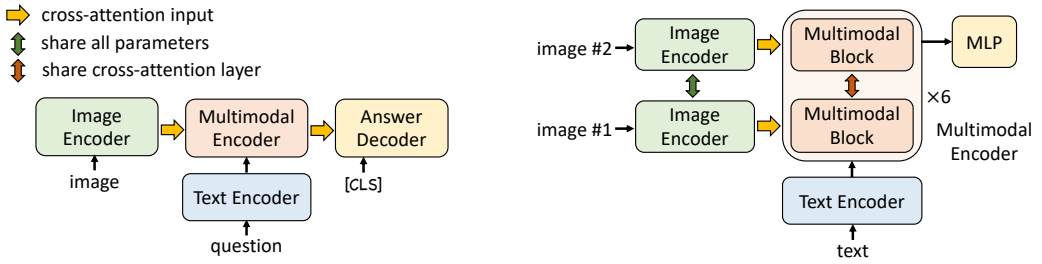


图 3：用于 VQA 和 NLVR^2 的模型架构。对于 VQA，我们附加一个自回归解码器，以根据图像-问题嵌入生成答案。对于 NLVR^2，我们在多模态编码器的每一层内复制 Transformer 块，以支持对两张图像进行推理。

答案解码器使用来自多模态编码器的预训练权重进行初始化，并通过条件语言建模损失进行微调。为了与现有方法进行公平比较，我们在推理过程中将解码器限制为只能从 3,192 个候选答案 [55] 中生成。

视觉推理的自然语言（NLVR^2 [19]）要求模型预测一段文本是否描述了一对图像。我们扩展了多模态编码器，使其能够对两张图像进行推理。如图 3b 所示，多模态编码器的每一层都被复制为两个连续的 Transformer 块，其中每个块包含一个自注意力层、一个交叉注意力层和一个前馈层（见图 1）。每层中的两个块使用相同的预训练权重进行初始化，并且两个交叉注意力层共享相同的键和值的线性投影权重。在训练过程中，这两个块接收图像对的两组图像嵌入。我们在多模态编码器的 [CLS] 表示上附加一个 MLP 分类器用于预测。

对于 NLVR^2，我们执行了一个额外的预训练步骤，以使新的多模态编码器能够对图像对进行编码。我们设计了一个文本分配（TA）任务，具体如下：给定一对图像和一段文本，模型需要将该文本分配给第一张图像、第二张图像，或两者都不分配。我们将其视为一个三分类问题，并在 [CLS] 表示上使用一个全连接层来预测分配结果。我们仅使用 4M 张图像（第 3.4 节）对 TA 预训练 1 个 epoch。

视觉定位旨在定位图像中与特定文本描述相对应的区域。我们研究弱监督设置，其中没有可用的边界框标注。我们在 RefCOCO+ [56] 数据集上进行实验，并遵循与图像-文本检索相同的策略，仅使用图像-文本监督对模型进行微调。在推理过程中，我们扩展 Grad-CAM [9] 以获取热力图，并使用它们对 [53] 提供的检测候选区域进行排序。

6 实验

6.1 对所提方法的评估

首先，我们评估所提出方法（*i.e.* 图像-文本对比学习、对比式困难负样本挖掘和动量蒸馏）的有效性。表 1 展示了我们方法不同变体在下游任务上的性能。与基线预训练任务（MLM+ ITM）相比，加入 ITC 显著提升了预训练模型在各方面的性能

#Pre-train Images	Training tasks	TR (flickr test)	IR (flickr test)	SNLI-VE (test)	NLVR^2 (test-P)	VQA (test-dev)
4M	MLM + ITM	93.96	88.55	77.06	77.51	71.40
	ITC + MLM + ITM	96.55	91.69	79.15	79.88	73.29
	ITC + MLM + ITM_{hard}	97.01	92.16	79.77	80.35	73.81
	ITC_{MoD} + MLM + ITM_{hard}	97.33	92.43	79.99	80.34	74.06
	Full (ITC_{MoD} + MLM_{MoD} + ITM_{hard})	97.47	92.58	80.12	80.44	74.42
	ALBEF (Full + $\text{MoD}_{\text{Downstream}}$)	97.83	92.65	80.30	80.50	74.54
14M	ALBEF	98.70	94.07	80.91	83.14	75.84

表 1：所提出方法在四个下游 V+L 任务上的评估。对于文本检索（TR）和图像检索（IR），我们报告 R@1、R@5 和 R@10 的平均值。ITC：图像-文本对比学习。MLM：掩码语言建模。ITM_{hard}：采用对比式困难负样本挖掘的图像-文本匹配。MoD：动量蒸馏。$\text{MoD}_{\text{Downstream}}$：下游任务上的动量蒸馏。

Method	# Pre-train Images	Flickr30K (1K test set)						MSCOCO (5K test set)					
		TR			IR			TR			IR		
		R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10
UNITER	4M	87.3	98.0	99.2	75.6	94.1	96.8	65.7	88.6	93.8	52.9	79.9	88.0
VILLA	4M	87.9	97.5	98.8	76.3	94.2	96.8	-	-	-	-	-	-
OSCAR	4M	-	-	-	-	-	-	70.0	91.1	95.5	54.0	80.8	88.5
ALIGN	1.2B	95.3	99.8	100.0	84.9	97.4	98.6	77.0	93.5	96.9	59.9	83.3	89.8
ALBEF	4M	94.3	99.4	99.8	82.8	96.7	98.4	73.1	91.4	96.0	56.8	81.5	89.2
ALBEF	14M	**95.9**	**99.8**	**100.0**	**85.6**	**97.5**	**98.9**	**77.6**	**94.3**	**97.2**	**60.7**	**84.3**	**90.5**

Table 2: Fine-tuned image-text retrieval results on Flickr30K and COCO datasets.

Method	# Pre-train Images	Flickr30K (1K test set)					
		TR			IR		
		R@1	R@5	R@10	R@1	R@5	R@10
UNITER [2]	4M	83.6	95.7	97.7	68.7	89.2	93.9
CLIP [6]	400M	88.0	98.7	99.4	68.7	90.6	95.2
ALIGN [7]	1.2B	88.6	98.7	99.7	75.7	93.8	96.8
ALBEF	4M	90.5	98.8	99.7	76.8	93.7	96.7
ALBEF	14M	**94.1**	**99.5**	**99.7**	**82.8**	**96.3**	**98.1**

Table 3: Zero-shot image-text retrieval results on Flickr30K.

Method	VQA		NLVR^2		SNLI-VE	
	test-dev	test-std	dev	test-P	val	test
VisualBERT [13]	70.80	71.00	67.40	67.00	-	-
VL-BERT [10]	71.16	-	-	-	-	-
LXMERT [1]	72.42	72.54	74.90	74.50	-	-
12-in-1 [12]	73.15	-	-	78.87	-	76.95
UNITER [2]	72.70	72.91	77.18	77.85	78.59	78.28
VL-BART/T5 [54]	-	71.3	-	73.6	-	-
ViLT [21]	70.94	-	75.24	76.21	-	-
OSCAR [3]	73.16	73.44	78.07	78.36	-	-
VILLA [8]	73.59	73.67	78.39	79.30	79.47	79.03
ALBEF (4M)	74.54	74.70	80.24	80.50	80.14	80.30
ALBEF (14M)	**75.84**	**76.04**	**82.55**	**83.14**	**80.80**	**80.91**

Table 4: Comparison with state-of-the-art methods on downstream vision-language tasks.

all tasks. The proposed hard negative mining improves ITM by finding more informative training samples. Furthermore, adding momentum distillation improves learning for both ITC (row 4), MLM (row 5), and on all downstream tasks (row 6). In the last row, we show that ALBEF can effectively leverage more noisy web data to improve the pre-training performance.

6.2 Evaluation on Image-Text Retrieval

Table 2 and Table 3 report results on fine-tuned and zero-shot image-text retrieval, respectively. Our ALBEF achieves state-of-the-art performance, outperforming CLIP [6] and ALIGN [7] which are trained on orders of magnitude larger datasets. Given the considerable amount of improvement of ALBEF when the number of training images increases from 4M to 14M, we hypothesize that it has potential to further grow by training on larger-scale web image-text pairs.

6.3 Evaluation on VQA, NLVR, and VE

Table 4 reports the comparison with existing methods on other V+L understanding tasks. With 4M pre-training images, ALBEF already achieves state-of-the-art performance. With 14M pre-training images, ALBEF substantially outperforms existing methods, including methods that additionally use object tags [3] or adversarial data augmentation [8]. Compared to VILLA [8], ALBEF achieves absolute improvements of 2.37% on VQA test-std, 3.84% on NLVR^2 test-P, and 1.88% on SNLI-VE test. Because ALBEF is detector-free and requires lower resolution images, it also enjoys much faster inference speed compared to most existing methods (>10 times faster than VILLA on NLVR^2).

6.4 Weakly-supervised Visual Grounding

Table 5 shows the results on RefCOCO+, where ALBEF substantially outperforms existing methods [57, 58] (which use weaker text embeddings). The ALBEF$_{\text{itc}}$ variant computes Grad-CAM

Method	# Pre-train Images	Flickr30K (1K test set)						MSCOCO (5K test set)					
		TR			IR			TR			IR		
		R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10
UNITER	4M	87.3	98.0	99.2	75.6	94.1	96.8	65.7	88.6	93.8	52.9	79.9	88.0
VILLA	4M	87.9	97.5	98.8	76.3	94.2	96.8	-	-	-	-	-	-
OSCAR	4M	-	-	-	-	-	-	70.0	91.1	95.5	54.0	80.8	88.5
ALIGN	1.2B	95.3	99.8	100.0	84.9	97.4	98.6	77.0	93.5	96.9	59.9	83.3	89.8
ALBEF	4M	94.3	99.4	99.8	82.8	96.7	98.4	73.1	91.4	96.0	56.8	81.5	89.2
ALBEF	14M	**95.9**	**99.8**	**100.0**	**85.6**	**97.5**	**98.9**	**77.6**	**94.3**	**97.2**	**60.7**	**84.3**	**90.5**

表2：在 Flickr30K 和 COCO 上微调后的图像-文本检索结果数据集。

Method	# Pre-train Images	Flickr30K (1K test set)					
		TR			IR		
		R@1	R@5	R@10	R@1	R@5	R@10
UNITER [2]	4M	83.6	95.7	97.7	68.7	89.2	93.9
CLIP [6]	400M	88.0	98.7	99.4	68.7	90.6	95.2
ALIGN [7]	1.2B	88.6	98.7	99.7	75.7	93.8	96.8
ALBEF	4M	90.5	98.8	99.7	76.8	93.7	96.7
ALBEF	14M	**94.1**	**99.5**	**99.7**	**82.8**	**96.3**	**98.1**

表 3：Flickr30K 上的零样本图文检索结果。

Method	VQA		NLVR^2		SNLI-VE	
	test-dev	test-std	dev	test-P	val	test
VisualBERT [13]	70.80	71.00	67.40	67.00	-	-
VL-BERT [10]	71.16	-	-	-	-	-
LXMERT [1]	72.42	72.54	74.90	74.50	-	-
12-in-1 [12]	73.15	-	-	78.87	-	76.95
UNITER [2]	72.70	72.91	77.18	77.85	78.59	78.28
VL-BART/T5 [54]	-	71.3	-	73.6	-	-
ViLT [21]	70.94	-	75.24	76.21	-	-
OSCAR [3]	73.16	73.44	78.07	78.36	-	-
VILLA [8]	73.59	73.67	78.39	79.30	79.47	79.03
ALBEF (4M)	74.54	74.70	80.24	80.50	80.14	80.30
ALBEF (14M)	**75.84**	**76.04**	**82.55**	**83.14**	**80.80**	**80.91**

表4：与下游视觉-语言任务上的最先进方法的比较。

所有任务。所提出的困难负样本挖掘通过找到更具信息量的训练样本来提升 ITM。此外，加入动量蒸馏能够改善 ITC（第 4 行）、MLM（第 5 行）以及所有下游任务（第 6 行）的学习效果。在最后一行，我们展示了 ALBEF 能够有效利用更多噪声较大的网络数据来提升预训练性能。

6.2 图像-文本检索评估

表2和表3分别报告了微调和零样本图像-文本检索的结果。我们的 ALBEF 达到了最先进的性能，优于在大几个数量级的数据集上训练的 CLIP [6] 和 ALIGN [7]。鉴于当训练图像数量从 4M 增加到 14M 时 ALBEF 取得了显著提升，我们假设通过在更大规模的网络图像-文本对上进行训练，它仍有进一步提升的潜力。

6.3 在 VQA、NLVR 和 VE 上的评估

表 4 报告了在其他 V+L 理解任务上与现有方法的比较。使用 4M 预训练图像时，ALBEF 已经达到了最先进的性能。使用 14M 预训练图像时，ALBEF 显著优于现有方法，包括额外使用目标标签 [3] 或对抗性数据增强 [8] 的方法。与 VILLA [8] 相比，ALBEF 在 VQA test-std 上取得了 2.37% 的绝对提升，在 NLVR^2 test-P 上取得了 3.84% 的绝对提升，并在 SNLI-VE test 上取得了 1.88% 的绝对提升。由于 ALBEF 无需检测器且需要较低分辨率的图像，因此与大多数现有方法相比，它还具有快得多的推理速度（在 NLVR^2 上比 VILLA 快 >10 倍）。

6.4 弱监督视觉定位

表5展示了在 RefCOCO+ 上的结果，其中 ALBEF 显著优于现有方法 [57, 58]（这些方法使用较弱的文本嵌入）。ALBEF$_{\text{itc}}$ 变体计算 Grad-CAM

Method	Val	TestA	TestB
ARN [57]	32.78	34.35	32.13
CCL [58]	34.29	36.91	33.56
$\text{ALBEF}_{\text{itc}}$	51.58	60.09	40.19
$\text{ALBEF}_{\text{itm}}$	**58.46**	**65.89**	**46.25**

Table 5: Weakly-supervised visual grounding on RefCOCO+ [56] dataset.



Figure 4: Grad-CAM visualization on the cross-attention maps in the 3rd layer of the multimodal encoder.

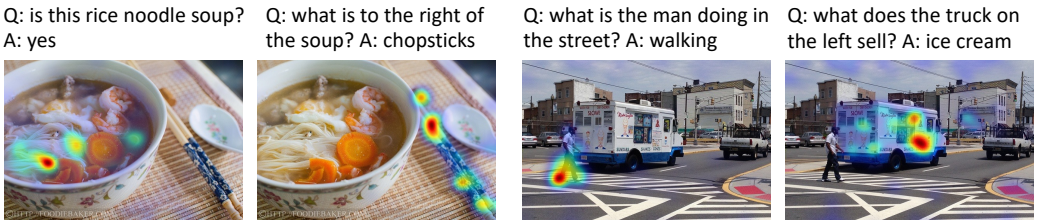


Figure 5: Grad-CAM visualizations on the cross-attention maps of the multimodal encoder for the VQA model.



Figure 6: Grad-CAM visualizations on the cross-attention maps corresponding to individual words.

visualizations on the self-attention maps in the last layer of the image encoder, where the gradients are acquired by maximizing the image-text similarity s_{itc}. The $\text{ALBEF}_{\text{itm}}$ variant computes Grad-CAM on the cross-attention maps in the 3rd layer of the multimodal encoder (which is a layer specialized in grounding), where the gradients are acquired by maximizing the image-text matching score s_{itm}. Figure 4 provides a few visualizations. More analysis is in Appendix.

We provide the Grad-CAM visualizations for VQA in Figure 5. As can be seen in Appendix, the Grad-CAM visualizations from ALBEF are highly correlated with where humans would look when making decisions. In Figure 6, we show per-word visualizations for COCO. Notice how our model not only grounds objects, but also their attributes and relationships.

6.5 Ablation Study

Table 6 studies the effect of various design choices on image-text retrieval. Since we use s_{itc} to filter top-k candidates during inference, we vary k and report its effect. In general, the ranking result acquired by s_{itm} is not sensitive to changes in k. We also validate the effect of hard negative mining in the last column.

Flickr30K	w/ hard negs				w/o hard negs
	s_{itc}	$k=16$	$k=128$	$k=256$	$k=128$
TR	97.30	98.60	98.57	98.57	98.22 (−0.35)
IR	90.95	93.64	93.99	93.95	93.68 (−0.31)

Table 6: Ablation study on fine-tuned image-text retrieval. The average recall on the test set is reported. We use s_{itc} to filter top-k candidates and calculate their s_{itm} score for ranking.

Table 7 studies the effect of text-assignment (TA) pre-training and parameter sharing on NLVR^2. We examine three strategies: (1) the two mutimodal blocks share all parameters, (2) only the cross-attention (CA) layers are shared, (3) no sharing. Without TA, sharing the entire block has better performance. With TA to pre-train the model for image-pair, sharing CA leads to the best performance.

NLVR^2	w/ TA			w/o TA		
	share all	share CA	no share	share all	share CA	no share
dev	82.13	82.55	81.93	80.52	80.28	77.84
test-P	82.36	83.14	82.85	81.29	80.45	77.58

Table 7: Ablation study on NLVR^2.

7 Conclusion and Social Impacts

This paper proposes ALBEF, a new framework for vision-language representation learning. ALBEF first aligns the unimodal image representation and text representation before fusing them with a multimodal encoder. We theoretically and experimentally verify the effectiveness of the proposed

Method	Val	TestA	TestB
ARN [57]	32.78	34.35	32.13
CCL [58]	34.29	36.91	33.56
$\text{ALBEF}_{\text{itc}}$	51.58	60.09	40.19
$\text{ALBEF}_{\text{itm}}$	**58.46**	**65.89**	**46.25**

表 5：在 RefCOCO+ [56] 数据集上的弱监督视觉定位。



图 4：多模态编码器第 3 层中交叉注意力图上的 Grad-CAM 可视化。

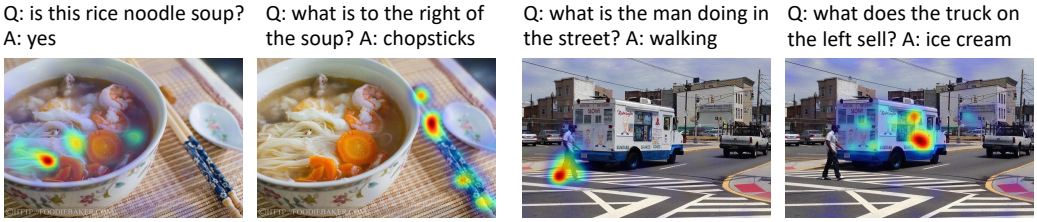


图 5：VQA 模型的多模态编码器中交叉注意力图上的 Grad-CAM 可视化。



图 6：Grad- 对应于 ind 的交叉注意力图上的 CAM 可视化 单个单词。

图像编码器最后一层的自注意力图上的可视化，其中梯度通过最大化图像-文本相似度 s_{itc} 获得。$\text{ALBEF}_{\text{itm}}$ 变体在多模态编码器第 3 层的交叉注意力图上计算 Grad-CAM（该层是专门用于定位的层），其中梯度通过最大化图像-文本匹配分数 s_{itm} 获得。图 4 提供了一些可视化结果。更多分析见附录。

我们在图 5 中提供了 VQA 的 Grad-CAM 可视化结果。如附录所示，ALBEF 的 Grad-CAM 可视化结果与人类在做出决策时的关注位置高度相关。在图 6 中，我们展示了 COCO 的逐词可视化结果。可以注意到，我们的模型不仅能定位物体，还能定位其属性和关系。

6.5 消融研究

表 6 研究了各种设计选择对图文检索的影响。由于我们在推理过程中使用 s_{itc} 来筛选 top-k 候选项，因此我们改变 k 并报告其影响。总体而言，由 s_{itm} 获得的排序结果对变化并不敏感在 k 中。我们还在最后的 c 中验证了困难负样本挖掘的效果 列。

Flickr30K	w/ hard negs				w/o hard negs
	s_{itc}	$k=16$	$k=128$	$k=256$	$k=128$
TR	97.30	98.60	98.57	98.57	98.22 (−0.35)
IR	90.95	93.64	93.99	93.95	93.68 (−0.31)

表 6：微调后的图文检索消融研究。报告了测试集上的平均召回率。我们使用 s_{itc} 筛选前 k 个候选项，并计算它们的 s_{itm} 分数用于排序。

表 7 研究了文本分配（TA）预训练和参数共享对 NLVR^2 的影响。我们考察了三种策略：（1）两个多模态块共享所有参数，（2）仅交叉-

NLVR^2	w/ TA			w/o TA		
	share all	share CA	no share	share all	share CA	no share
dev	82.13	82.55	81.93	80.52	80.28	77.84
test-P	82.36	83.14	82.85	81.29	80.45	77.58

表 7：NLVR^2 上的消融研究。

注意力（CA）层共享，（3）不共享。没有 TA 时，共享整个块的性能更好。使用 TA 对图像对模型进行预训练时，共享 CA 可获得最佳性能。

7 结论与社会影响

本文提出了 ALBEF，这是一种用于视觉-语言表征学习的新框架。ALBEF 首先对齐单模态图像表征和文本表征，然后使用多模态编码器将它们融合。我们从理论和实验上验证了所提出方法的有效性

image-text contrastive learning and momentum distillation. Compared to existing methods, ALBEF offers better performance and faster inference speed on multiple downstream V+L tasks.

While our paper shows promising results on vision-language representation learning, additional analysis on the data and the model is necessary before deploying it in practice, because web data may contain unintended private information, unsuitable images, or harmful texts, and only optimizing accuracy may have unwanted social implications.

References

[1] Tan, H., M. Bansal. LXMERT: learning cross-modality encoder representations from transformers. In K. Inui, J. Jiang, V. Ng, X. Wan, eds., *EMNLP*, pages 5099–5110. 2019.

[2] Chen, Y., L. Li, L. Yu, et al. UNITER: universal image-text representation learning. In *ECCV*, vol. 12375, pages 104–120. 2020.

[3] Li, X., X. Yin, C. Li, et al. Oscar: Object-semantics aligned pre-training for vision-language tasks. In *ECCV*, pages 121–137. 2020.

[4] Sharma, P., N. Ding, S. Goodman, et al. Conceptual captions: A cleaned, hypernymed, image alt-text dataset for automatic image captioning. In I. Gurevych, Y. Miyao, eds., *ACL*, pages 2556–2565. 2018.

[5] Ordonez, V., G. Kulkarni, T. L. Berg. Im2text: Describing images using 1 million captioned photographs. In J. Shawe-Taylor, R. S. Zemel, P. L. Bartlett, F. C. N. Pereira, K. Q. Weinberger, eds., *NIPS*, pages 1143–1151. 2011.

[6] Radford, A., J. W. Kim, C. Hallacy, et al. Learning transferable visual models from natural language supervision. *arXiv preprint arXiv:2103.00020*, 2021.

[7] Jia, C., Y. Yang, Y. Xia, et al. Scaling up visual and vision-language representation learning with noisy text supervision. *arXiv preprint arXiv:2102.05918*, 2021.

[8] Gan, Z., Y. Chen, L. Li, et al. Large-scale adversarial training for vision-and-language representation learning. In H. Larochelle, M. Ranzato, R. Hadsell, M. Balcan, H. Lin, eds., *NeurIPS*. 2020.

[9] Selvaraju, R. R., M. Cogswell, A. Das, et al. Grad-cam: Visual explanations from deep networks via gradient-based localization. In *ICCV*, pages 618–626. 2017.

[10] Su, W., X. Zhu, Y. Cao, et al. Vl-bert: Pre-training of generic visual-linguistic representations. In *ICLR*. 2020.

[11] Lu, J., D. Batra, D. Parikh, et al. Vilbert: Pretraining task-agnostic visiolinguistic representations for vision-and-language tasks. In H. M. Wallach, H. Larochelle, A. Beygelzimer, F. d'Alché-Buc, E. B. Fox, R. Garnett, eds., *NeurIPS*, pages 13–23. 2019.

[12] Lu, J., V. Goswami, M. Rohrbach, et al. 12-in-1: Multi-task vision and language representation learning. In *CVPR*, pages 10434–10443. 2020.

[13] Li, L. H., M. Yatskar, D. Yin, et al. Visualbert: A simple and performant baseline for vision and language. *arXiv preprint arXiv:1908.03557*, abs/1908.03557, 2019.

[14] Qi, D., L. Su, J. Song, et al. Imagebert: Cross-modal pre-training with large-scale weak-supervised image-text data. *arXiv preprint arXiv:2001.07966*, 2020.

[15] Li, G., N. Duan, Y. Fang, et al. Unicoder-vl: A universal encoder for vision and language by cross-modal pre-training. In *AAAI*, pages 11336–11344. 2020.

[16] Yu, F., J. Tang, W. Yin, et al. Ernie-vil: Knowledge enhanced vision-language representations through scene graph. *arXiv preprint arXiv:2006.16934*, 2020.

[17] Zhang, P., X. Li, X. Hu, et al. Vinvl: Making visual representations matter in vision-language models. *arXiv preprint arXiv:2101.00529*, 2021.

[18] Huang, Z., Z. Zeng, Y. Huang, et al. Seeing out of the box: End-to-end pre-training for vision-language representation learning. *arXiv preprint arXiv:2104.03135*, 2021.

[19] Suhr, A., S. Zhou, A. Zhang, et al. A corpus for reasoning about natural language grounded in photographs. In A. Korhonen, D. R. Traum, L. Màrquez, eds., *ACL*, pages 6418–6428. 2019.

图文对比学习和动量蒸馏。与现有方法相比，ALBEF 在多个下游 V+L 任务上提供了更好的性能和更快的推理速度。

尽管我们的论文在视觉-语言表征学习方面展示了有前景的结果，但在实际部署之前，仍有必要对数据和模型进行额外分析，因为网络数据可能包含无意泄露的私人信息、不适宜的图像或有害文本，而仅优化准确率可能会带来不良的社会影响。

参考文献

[1] Tan, H., M. Bansal. LXMERT：从 Transformer 中学习跨模态编码器表示。载于 K. Inui, J. Jiang, V. Ng, X. Wan 编，*EMNLP*，第 5099–5110 页。2019。 [2] Chen, Y., L. Li, L. Yu, 等。UNITER：通用图像-文本表示学习。载于 *ECCV*，第 12375 卷，第 104–120 页。2020。 [3] Li, X., X. Yin, C. Li, 等。Oscar：用于视觉-语言任务的对象语义对齐预训练。载于 *ECCV*，第 121–137 页。2020。 [4] Sharma, P., N. Ding, S. Goodman, 等。Conceptual Captions：一个用于自动图像描述生成的清洗过、上位词化的图像替代文本数据集。载于 I. Gurevych, Y. Miyao 编，*ACL*，第 2556–2565 页。2018。 [5] Ordonez, V., G. Kulkarni, T. L. Berg. Im2text：使用 100 万张带说明的照片描述图像。载于 J. Shawe-Taylor, R. S. Zemel, P. L. Bartlett, F. C. N. Pereira, K. Q. Weinberger 编，*NIPS*，第 1143–1151 页。2011。 [6] Radford, A., J. W. Kim, C. Hallacy, 等。从自然语言监督中学习可迁移的视觉模型。*arXiv preprint arXiv:2103.00020*，2021。 [7] Jia, C., Y. Yang, Y. Xia, 等。利用噪声文本监督扩展视觉与视觉-语言表示学习。*arXiv preprint arXiv:2102.05918*，2021。 [8] Gan, Z., Y. Chen, L. Li, 等。用于视觉与语言表示学习的大规模对抗训练。载于 H. Larochelle, M. Ranzato, R. Hadsell, M. Balcan, H. Lin 编，*NeurIPS*。2020。 [9] Selvaraju, R. R., M. Cogswell, A. Das, 等。Grad-cam：通过基于梯度的定位从深度网络中获得视觉解释。载于 *ICCV*，第 618–626 页。2017。 [10] Su, W., X. Zhu, Y. Cao, 等。Vl-bert：通用视觉-语言表示的预训练。载于 *ICLR*。2020。 [11] Lu, J., D. Batra, D. Parikh, 等。Vilbert：面向视觉与语言任务的任务无关视觉语言表示预训练。载于 H. M. Wallach, H. Larochelle, A. Beygelzimer, F. d’Alché-Buc, E. B. Fox, R. Garnett 编，*NeurIPS*，第 13–23 页。2019。 [12] Lu, J., V. Goswami, M. Rohrbach, 等。12-in-1：多任务视觉与语言表示学习。载于 *CVPR*，第 10434–10443 页。2020。 [13] Li, L. H., M. Yatskar, D. Yin, 等。Visualbert：一种用于视觉与语言的简单且高性能的基线。*arXiv preprint arXiv:1908.03557*，abs/1908.03557，2019。 [14] Qi, D., L. Su, J. Song, 等。Imagebert：利用大规模弱监督图像-文本数据进行跨模态预训练。*arXiv preprint arXiv:2001.07966*，2020。 [15] Li, G., N. Duan, Y. Fang, 等。Unicoder-vl：通过跨模态预训练构建的视觉与语言通用编码器。载于 *AAAI*，第 11336–11344 页。2020。 [16] Yu, F., J. Tang, W. Yin, 等。Ernie-vil：通过场景图增强知识的视觉-语言表示。*arXiv preprint arXiv:2006.16934*，2020。 [17] Zhang, P., X. Li, X. Hu, 等。Vinvl：让视觉表示在视觉-语言模型中发挥作用。*arXiv preprint arXiv:2101.00529*，2021。 [18] Huang, Z., Z. Zeng, Y. Huang, 等。跳出框外看世界：用于视觉-语言表示学习的端到端预训练。*arXiv preprint arXiv:2104.03135*，2021。 [19] Suhr, A., S. Zhou, A. Zhang, 等。用于推理基于照片的自然语言的语料库。载于 A. Korhonen, D. R. Traum, L. Màrquez 编，*ACL*，第 6418–6428 页。2019。

[20] Antol, S., A. Agrawal, J. Lu, et al. VQA: visual question answering. In *ICCV*, pages 2425–2433. 2015.

[21] Kim, W., B. Son, I. Kim. Vilt: Vision-and-language transformer without convolution or region supervision. *arXiv preprint arXiv:2102.03334*, 2021.

[22] Faghri, F., D. J. Fleet, J. R. Kiros, et al. VSE++: improving visual-semantic embeddings with hard negatives. In *BMVC*, page 12. 2018.

[23] Li, K., Y. Zhang, K. Li, et al. Visual semantic reasoning for image-text matching. In *ICCV*, pages 4653–4661. 2019.

[24] He, K., H. Fan, Y. Wu, et al. Momentum contrast for unsupervised visual representation learning. In *CVPR*. 2020.

[25] Chen, T., S. Kornblith, M. Norouzi, et al. A simple framework for contrastive learning of visual representations. In *ICML*. 2020.

[26] Li, J., P. Zhou, C. Xiong, et al. Prototypical contrastive learning of unsupervised representations. In *ICLR*. 2021.

[27] Li, J., C. Xiong, S. C. Hoi. Mopro: Webly supervised learning with momentum prototypes. In *ICLR*. 2021.

[28] Hinton, G., O. Vinyals, J. Dean. Distilling the knowledge in a neural network. *arXiv preprint arXiv:1503.02531*, 2015.

[29] Zagoruyko, S., N. Komodakis. Paying more attention to attention: Improving the performance of convolutional neural networks via attention transfer. In *ICLR*. 2017.

[30] Furlanello, T., Z. C. Lipton, M. Tschannen, et al. Born-again neural networks. In J. G. Dy, A. Krause, eds., *ICML*, pages 1602–1611. 2018.

[31] Touvron, H., M. Cord, M. Douze, et al. Training data-efficient image transformers & distillation through attention. *arXiv preprint arXiv:2012.12877*, 2020.

[32] Sanh, V., L. Debut, J. Chaumond, et al. Distilbert, a distilled version of bert: smaller, faster, cheaper and lighter. *arXiv preprint arXiv:1910.01108*, 2019.

[33] Zhang, Y., T. Xiang, T. M. Hospedales, et al. Deep mutual learning. In *CVPR*, pages 4320–4328. 2018.

[34] Anil, R., G. Pereyra, A. Passos, et al. Large scale distributed neural network training through online distillation. In *ICLR*. 2018.

[35] Tarvainen, A., H. Valpola. Mean teachers are better role models: Weight-averaged consistency targets improve semi-supervised deep learning results. In *NIPS*, pages 1195–1204. 2017.

[36] Li, J., R. Socher, S. C. Hoi. Dividemix: Learning with noisy labels as semi-supervised learning. In *ICLR*. 2020.

[37] Cheng, R., B. Wu, P. Zhang, et al. Data-efficient language-supervised zero-shot learning with self-distillation. *arXiv preprint arXiv:2104.08945*, 2021.

[38] Dosovitskiy, A., L. Beyer, A. Kolesnikov, et al. An image is worth 16x16 words: Transformers for image recognition at scale. In *ICLR*. 2021.

[39] Vaswani, A., N. Shazeer, N. Parmar, et al. Attention is all you need. In I. Guyon, U. von Luxburg, S. Bengio, H. M. Wallach, R. Fergus, S. V. N. Vishwanathan, R. Garnett, eds., *NIPS*, pages 5998–6008. 2017.

[40] Devlin, J., M. Chang, K. Lee, et al. BERT: pre-training of deep bidirectional transformers for language understanding. In J. Burstein, C. Doran, T. Solorio, eds., *NAACL*, pages 4171–4186. 2019.

[41] Lin, T., M. Maire, S. J. Belongie, et al. Microsoft COCO: common objects in context. In D. J. Fleet, T. Pajdla, B. Schiele, T. Tuytelaars, eds., *ECCV*, vol. 8693, pages 740–755. 2014.

[42] Krishna, R., Y. Zhu, O. Groth, et al. Visual genome: Connecting language and vision using crowdsourced dense image annotations. *IJCV*, 123(1):32–73, 2017.

[43] Changpinyo, S., P. Sharma, N. Ding, et al. Conceptual 12M: Pushing web-scale image-text pre-training to recognize long-tail visual concepts. In *CVPR*. 2021.

[20] Antol, S., A. Agrawal, J. Lu, 等. VQA：视觉问答。见 *ICCV*，第 2425–2433 页。2015。 [21] Kim, W., B. Son, I. Kim. Vilt：无需卷积或区域监督的视觉-语言 Transformer。*arXiv preprint arXiv:2102.03334*，2021。 [22] Faghri, F., D. J. Fleet, J. R. Kiros, 等. VSE++：通过困难负样本改进视觉-语义嵌入。见 *BMVC*，第 12 页。2018。 [23] Li, K., Y. Zhang, K. Li, 等. 用于图像-文本匹配的视觉语义推理。见 *ICCV*，第 4653–4661 页。2019。 [24] He, K., H. Fan, Y. Wu, 等. 用于无监督视觉表征学习的动量对比。见 *CVPR*。2020。 [25] Chen, T., S. Kornblith, M. Norouzi, 等. 一种用于视觉表征对比学习的简单框架。见 *ICML*。2020。 [26] Li, J., P. Zhou, C. Xiong, 等. 无监督表征的原型对比学习。见 *ICLR*。2021。 [27] Li, J., C. Xiong, S. C. Hoi. Mopro：具有动量原型的网络监督学习。见 *ICLR*。2021。 [28] Hinton, G., O. Vinyals, J. Dean. 提炼神经网络中的知识。*arXiv preprint arXiv:1503.02531*，2015。 [29] Zagoruyko, S., N. Komodakis. 更加关注注意力：通过注意力迁移提高卷积神经网络的性能。见 *ICLR*。2017。 [30] Furlanello, T., Z. C. Lipton, M. Tschannen, 等. 重生神经网络。见 J. G. Dy, A. Krause 编，*ICML*，第 1602–1611 页。2018。 [31] Touvron, H., M. Cord, M. Douze, 等. 通过注意力训练数据高效的图像 Transformer 与蒸馏。*arXiv preprint arXiv:2012.12877*，2020。 [32] Sanh, V., L. Debut, J. Chaumond, 等. Distilbert，bert 的蒸馏版本：更小、更快、更便宜且更轻量。*arXiv preprint arXiv:1910.01108*，2019。 [33] Zhang, Y., T. Xiang, T. M. Hospedales, 等. 深度互学习。见 *CVPR*，第 4320–4328 页。2018。 [34] Anil, R., G. Pereyra, A. Passos, 等. 通过在线蒸馏进行大规模分布式神经网络训练。见 *ICLR*。2018。 [35] Tarvainen, A., H. Valpola. 平均教师是更好的榜样：权重平均的一致性目标改进半监督深度学习结果。见 *NIPS*，第 1195–1204 页。2017。 [36] Li, J., R. Socher, S. C. Hoi. Dividemix：将带噪标签学习视为半监督学习。见 *ICLR*。2020。 [37] Cheng, R., B. Wu, P. Zhang, 等. 通过自蒸馏实现数据高效的语言监督零样本学习。*arXiv preprint arXiv:2104.08945*，2021。 [38] Dosovitskiy, A., L. Beyer, A. Kolesnikov, 等. 一张图像抵得上 16x16 个词：用于大规模图像识别的 Transformer。见 *ICLR*。2021。 [39] Vaswani, A., N. Shazeer, N. Parmar, 等. 注意力就是你所需要的一切。见 I. Guyon, U. von Luxburg, S. Bengio, H. M. Wallach, R. Fergus, S. V. N. Vishwanathan, R. Garnett 编，*NIPS*，第 5998–6008 页。2017。 [40] Devlin, J., M. Chang, K. Lee, 等. BERT：用于语言理解的深度双向 Transformer 预训练。见 J. Burstein, C. Doran, T. Solorio 编，*NAACL*，第 4171–4186 页。2019。 [41] Lin, T., M. Maire, S. J. Belongie, 等. Microsoft COCO：上下文中的常见物体。见 D. J. Fleet, T. Pajdla, B. Schiele, T. Tuytelaars 编，*ECCV*，第 8693 卷，第 740–755 页。2014。 [42] Krishna, R., Y. Zhu, O. Groth, 等. Visual genome：使用众包密集图像标注连接语言与视觉。*IJCV*，123(1):32–73，2017。 [43] Changpinyo, S., P. Sharma, N. Ding, 等. Conceptual 12M：推动网络规模图像-文本预训练以识别长尾视觉概念。见 *CVPR*。2021。

[44] Loshchilov, I., F. Hutter. Decoupled weight decay regularization. *arXiv preprint arXiv:1711.05101*, 2017.

[45] Cubuk, E. D., B. Zoph, J. Shlens, et al. Randaugment: Practical automated data augmentation with a reduced search space. In *CVPR Workshops*, pages 702–703. 2020.

[46] Tian, Y., C. Sun, B. Poole, et al. What makes for good views for contrastive learning? In H. Larochelle, M. Ranzato, R. Hadsell, M. Balcan, H. Lin, eds., *NeurIPS*. 2020.

[47] Oord, A. v. d., Y. Li, O. Vinyals. Representation learning with contrastive predictive coding. *arXiv preprint arXiv:1807.03748*, 2018.

[48] Kong, L., C. de Masson d'Autume, L. Yu, et al. A mutual information maximization perspective of language representation learning. In *ICLR*. OpenReview.net, 2020.

[49] Plummer, B. A., L. Wang, C. M. Cervantes, et al. Flickr30k entities: Collecting region-to-phrase correspondences for richer image-to-sentence models. In *ICCV*, pages 2641–2649. 2015.

[50] Do, V., O.-M. Camburu, Z. Akata, et al. e-snli-ve: Corrected visual-textual entailment with natural language explanations. *arXiv preprint arXiv:2004.03744*, 2020.

[51] Xie, N., F. Lai, D. Doran, et al. Visual entailment: A novel task for fine-grained image understanding. *arXiv preprint arXiv:1901.06706*, 2019.

[52] Goyal, Y., T. Khot, D. Summers-Stay, et al. Making the V in VQA matter: Elevating the role of image understanding in visual question answering. In *CVPR*, pages 6325–6334. 2017.

[53] Yu, L., Z. Lin, X. Shen, et al. Mattnet: Modular attention network for referring expression comprehension. In *CVPR*, pages 1307–1315. 2018.

[54] Cho, J., J. Lei, H. Tan, et al. Unifying vision-and-language tasks via text generation. *arXiv preprint arXiv:2102.02779*, 2021.

[55] Kim, J., J. Jun, B. Zhang. Bilinear attention networks. In S. Bengio, H. M. Wallach, H. Larochelle, K. Grauman, N. Cesa-Bianchi, R. Garnett, eds., *NIPS*, pages 1571–1581. 2018.

[56] Yu, L., P. Poirson, S. Yang, et al. Modeling context in referring expressions. In B. Leibe, J. Matas, N. Sebe, M. Welling, eds., *ECCV*, pages 69–85. 2016.

[57] Liu, X., L. Li, S. Wang, et al. Adaptive reconstruction network for weakly supervised referring expression grounding. In *ICCV*, pages 2611–2620. 2019.

[58] Zhang, Z., Z. Zhao, Z. Lin, et al. Counterfactual contrastive learning fo weakly-supervised vision-language grounding. In H. Larochelle, M. Ranzato, R. Hadsell, M. Balcan, H. Lin, eds., *NeurIPS*. 2020.

[59] Karpathy, A., F. Li. Deep visual-semantic alignments for generating image descriptions. In *CVPR*, pages 3128–3137. 2015.

[60] Bowman, S. R., G. Angeli, C. Potts, et al. A large annotated corpus for learning natural language inference. In L. Màrquez, C. Callison-Burch, J. Su, D. Pighin, Y. Marton, eds., *EMNLP*, pages 632–642. 2015.

[61] Yu, Z., J. Yu, Y. Cui, et al. Deep modular co-attention networks for visual question answering. In *CVPR*, pages 6281–6290. 2019.

[62] Kazemzadeh, S., V. Ordonez, M. Matten, et al. Referitgame: Referring to objects in photographs of natural scenes. In A. Moschitti, B. Pang, W. Daelemans, eds., *EMNLP*. 2014.

[63] Das, A., H. Agrawal, C. L. Zitnick, et al. Human Attention in Visual Question Answering: Do Humans and Deep Networks Look at the Same Regions? 2016.

[44] Loshchilov, I., F. Hutter. 解耦权重衰减正则化。*arXiv preprint arXiv:1711.05101*, 2017. [45] Cubuk, E. D., B. Zoph, J. Shlens, 等. Randaugment：具有缩小搜索空间的实用自动化数据增强。在 *CVPR Workshops* 中，页 702–703。2020. [46] Tian, Y., C. Sun, B. Poole, 等. 什么样的视图有利于对比学习？在 H. Larochelle, M. Ranzato, R. Hadsell, M. Balcan, H. Lin 编，*NeurIPS* 中。2020. [47] Oord, A. v. d., Y. Li, O. Vinyals. 使用对比预测编码进行表示学习。*arXiv preprint arXiv:1807.03748*, 2018. [48] Kong, L., C. de Masson d’ Autume, L. Yu, 等. 语言表示学习的互信息最大化视角。在 *ICLR* 中。OpenReview.net, 2020. [49] Plummer, B. A., L. Wang, C. M. Cervantes, 等. Flickr30k 实体：为更丰富的图像到句子模型收集区域到短语的对应关系。在 *ICCV* 中，页 2641–2649。2015. [50] Do, V., O.-M. Camburu, Z. Akata, 等. e-snli-ve：带自然语言解释的修正版视觉-文本蕴含。*arXiv preprint arXiv:2004.03744*, 2020. [51] Xie, N., F. Lai, D. Doran, 等. 视觉蕴含：用于细粒度图像理解的一项新任务。*arXiv preprint arXiv:1901.06706*, 2019. [52] Goyal, Y., T. Khot, D. Summers-Stay, 等. 让 VQA 中的 V 发挥作用：提升图像理解在视觉问答中的角色。在 *CVPR* 中，页 6325–6334。2017. [53] Yu, L., Z. Lin, X. Shen, 等. Mattnet：用于指代表达理解的模块化注意力网络。在 *CVPR* 中，页 1307–1315。2018. [54] Cho, J., J. Lei, H. Tan, 等. 通过文本生成统一视觉-语言任务。*arXiv preprint arXiv:2102.02779*, 2021. [55] Kim, J., J. Jun, B. Zhang. 双线性注意力网络。在 S. Bengio, H. M. Wallach, H. Larochelle, K. Grauman, N. Cesa-Bianchi, R. Garnett 编，*NIPS* 中，页 1571–1581。2018. [56] Yu, L., P. Poirson, S. Yang, 等. 指代表达中的上下文建模。在 B. Leibe, J. Matas, N. Sebe, M. Welling 编，*ECCV* 中，页 69–85。2016. [57] Liu, X., L. Li, S. Wang, 等. 用于弱监督指代表达定位的自适应重构网络。在 *ICCV* 中，页 2611–2620。2019. [58] Zhang, Z., Z. Zhao, Z. Lin, 等. 用于弱监督视觉-语言定位的反事实对比学习。在 H. Larochelle, M. Ranzato, R. Hadsell, M. Balcan, H. Lin 编，*NeurIPS* 中。2020. [59] Karpathy, A., F. Li. 用于生成图像描述的深度视觉-语义对齐。在 *CVPR* 中，页 3128–3137。2015. [60] Bowman, S. R., G. Angeli, C. Potts, 等. 用于学习自然语言推理的大规模标注语料库。在 L. Màrquez, C. Callison-Burch, J. Su, D. Pighin, Y. Marton 编，*EMNLP* 中，页 632–642。2015. [61] Yu, Z., J. Yu, Y. Cui, 等. 用于视觉问答的深度模块化协同注意力网络。在 *CVPR* 中，页 6281–6290。2019. [62] Kazemzadeh, S., V. Ordonez, M. Matten, 等. Referitgame：在自然场景照片中指代物体。在 A. Moschitti, B. Pang, W. Daelemans 编，*EMNLP* 中。2014. [63] Das, A., H. Agrawal, C. L. Zitnick, 等. 视觉问答中的人类注意力：人类和深度网络看的是相同区域吗？2016.

A Downstream Task Details

Here we describe the implementation details for fine-tuning the pre-trained model. For all downstream tasks, we use the same RandAugment, AdamW optimizer, cosine learning rate decay, weight decay, and distillation weight as during pre-training. All downstream tasks receive input images of resolution 384×384. During inference, we resize the images without any cropping.

Image-Text Retrieval. We consider two datasets for this task: COCO and Flickr30K. We adopt the widely used Karpathy split [59] for both datasets. COCO contains 113/5k/5k for train/validation/test. Flickr30K contains 29k/1k/1k images for train/validation/test. We fine-tune for 10 epochs. The batch size is 256 and the initial learning rate is $1e^{-5}$.

Visual Entailment. We evaluate on the SNLI-VE dataset [51], which is constructed using the Stanford Natural Language Inference (SNLI) [60] and Flickr30K datasets. We follow the original dataset split with 29.8k images for training, 1k for evaluation, and 1k for test. We fine-tune the pre-trained model for 5 epochs with a batch size of 256 and an initial learning rate of $2e^{-5}$.

VQA. We conduct experiment on the VQA2.0 dataset [52], which is constructed using images from COCO. It contains 83k images for training, 41k for validation, and 81k for test. We report performance on the test-dev and test-std splits. Following most existing works [1, 2, 61], we use both training and validation sets for training, and include additional question-answer pairs from Visual Genome. Because many questions in the VQA dataset contains multiple answers, we weight the loss for each answer by its percentage of occurrence among all answers. We fine-tune the model for 8 epochs, using a batch size of 256 and an initial learning rate of $2e^{-5}$.

NLVR^2. We conduct experiments following the original train/val/test split in [19]. We fine-tune the model for 10 epochs, using a batch size of 128 and an initial learning rate of $2e^{-5}$. Because NLVR receives two input images, we perform an additional step of pre-training with text-assignment (TA) to prepare the model for reasoning over two images. The TA pre-training uses images of size 256×256. We pre-train for 1 epoch on the 4M dataset, using a batch size of 256 and a learning rate of $2e^{-5}$.

Visual Grounding. We conduct experiments on the RefCOCO+ dataset [56], which is collected using a two-player ReferitGame [62]. It contains 141,564 expressions for 19,992 images from COCO training set. Strictly speaking, our model is not allowed to see the val/test images of RefCOCO+, but it has been exposed to those images during pre-training. We hypothesize that this has little effect because these images only occupy a very small portion of the entire 14M pre-training images, and leave it as future work to decontaminate the data. During weakly-supervised fine-tuning, we follow the same strategy as image-text retrieval except that we do not perform random cropping, and train the model for 5 epochs. During inference, we use either s_{itc} or s_{itm} to compute the importance score for each 16×16 image patch. For ITC, we compute Grad-CAM visualizations on the self-attention maps *w.r.t* the `[CLS]` token in the last layer of the visual encoder, and average the heatmaps across all attention heads. For ITM, we compute Grad-CAM on the cross-attention maps in the 3rd layer of the multimodal encoder, and average them scores across all attention heads and all input text tokens. Quantitative comparison between ITC and ITM is shown in Table 5. Figure 7 shows the qualitative comparison. Since the multimodal encoder can better model image-text interactions, it produces better heatmaps that capture finer-grained details. In Figure 8, we report the grounding accuracy for each cross-attention layer and each individual attention head within the best-performing layer.

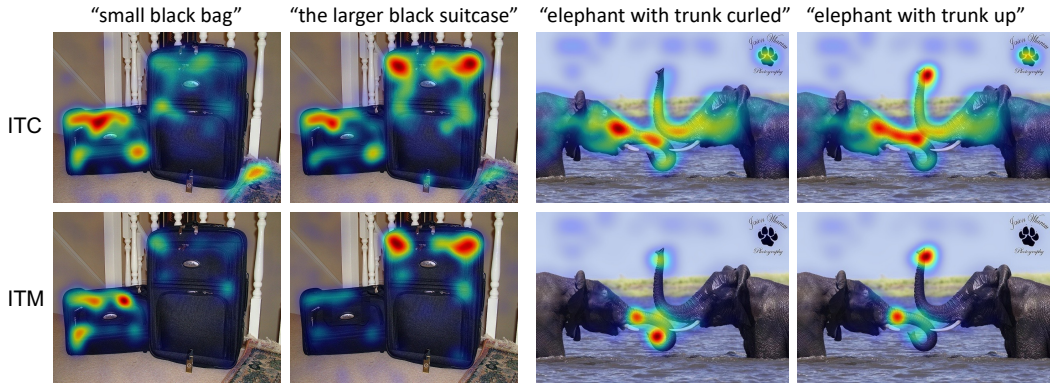


Figure 7: Grad-CAMs from the multimodal encoder capture finer-grained details such as "larger" and "curled".

下游任务详情

这里我们描述微调预训练模型的实现细节。对于所有下游任务，我们使用与预训练期间相同的 RandAugment、AdamW 优化器、余弦学习率衰减、权重衰减和蒸馏权重。所有下游任务都接收分辨率为 384 × 384 的输入图像。在推理期间，我们仅调整图像大小，不进行任何裁剪。

图文检索。我们针对该任务考虑两个数据集：COCO 和 Flickr30K。对于这两个数据集，我们均采用广泛使用的 Karpathy 划分 [59]。COCO 的训练/验证/测试集分别包含 113k/5k/5k 张图像。Flickr30K 的训练/验证/测试集分别包含 29k/1k/1k 张图像。我们微调 10 个 epoch。批大小为 256，初始学习率为 $1e^{-5}$。

视觉蕴含。我们在 SNLI-VE 数据集 [51] 上进行评估，该数据集是使用斯坦福自然语言推理（SNLI）[60] 和 Flickr30K 数据集构建的。我们遵循原始数据集划分，其中 29.8k 张图像用于训练，1k 张用于评估，1k 张用于测试。我们以 256 的批量大小和 $2e^{-5}$ 的初始学习率对预训练模型进行 5 个 epoch 的微调。

VQA。我们在 VQA2.0 数据集 [52] 上进行实验，该数据集使用 COCO 中的图像构建。它包含 83k 张训练图像、41k 张验证图像和 81k 张测试图像。我们报告 test-dev 和 test-std 划分上的性能。遵循大多数已有工作 [1, 2, 61]，我们同时使用训练集和验证集进行训练，并包含来自 Visual Genome 的额外问答对。由于 VQA 数据集中的许多问题包含多个答案，我们根据每个答案在所有答案中出现的百分比对其损失进行加权。我们对模型进行 8 个 epoch 的微调，批量大小为 256，初始学习率为 $2e^{-5}$。

NLVR^2。我们按照 [19] 中原始的训练/验证/测试划分进行实验。我们对模型进行 10 个 epoch 的微调，批量大小为 128，初始学习率为 $2e^{-5}$。由于 NLVR 接收两张输入图像，我们额外进行一步带有文本分配（TA）的预训练，以使模型能够对两张图像进行推理。TA 预训练使用尺寸为 256 × 256 的图像。我们在 4M 数据集上预训练 1 个 epoch，批量大小为 256，学习率为 $2e^{-5}$。

视觉定位。我们在 RefCOCO+ 数据集 [56] 上进行实验，该数据集使用双人 ReferitGame [62] 收集。它包含来自 COCO 训练集 19,992 张图像的 141,564 个表达。严格来说，我们的模型不允许看到 RefCOCO+ 的验证/测试图像，但在预训练期间它已经接触过这些图像。我们假设这影响很小，因为这些图像仅占全部 14M 预训练图像中非常小的一部分，并将数据去污染留作未来工作。在弱监督微调期间，我们遵循与图文检索相同的策略，只是不进行随机裁剪，并训练模型 5 个 epoch。在推理期间，我们使用 s_{itc} 或 s_{itm} 来计算每个 16 × 16 图像块的重要性分数。对于 ITC，我们在视觉编码器最后一层中 [CLS] 标记的自注意力图 *w.r.t* 上计算 Grad-CAM 可视化，并对所有注意力头的热力图取平均。对于 ITM，我们在多模态编码器第 3 层的交叉注意力图上计算 Grad-CAM，并对所有注意力头和所有输入文本标记的分数取平均。ITC 和 ITM 之间的定量比较如表 5 所示。图 7 展示了定性比较。由于多模态编码器能够更好地建模图文交互，因此它生成了更好的热力图，能够捕捉更细粒度的细节。在图 8 中，我们报告了每个交叉注意力层以及表现最佳层中每个单独注意力头的定位准确率。

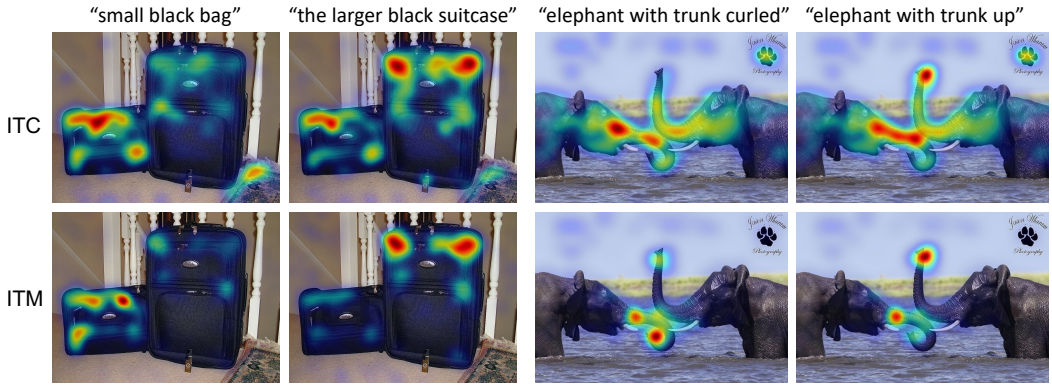


图 7：来自多模态编码器的 Grad-CAM 捕捉到了更细粒度的细节，例如“更大”和“卷曲”。

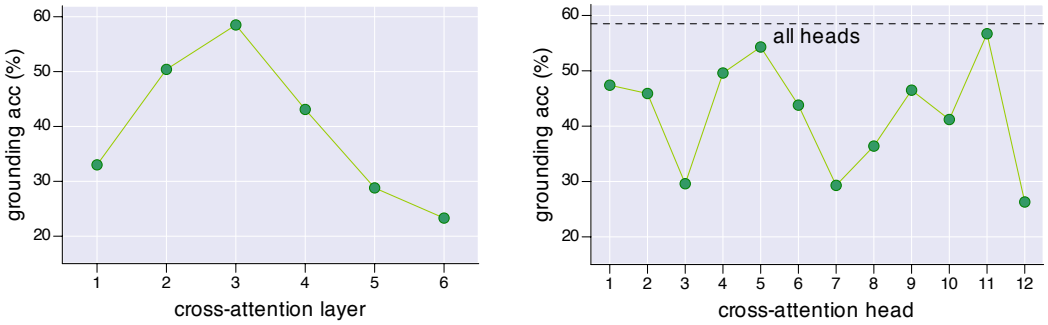


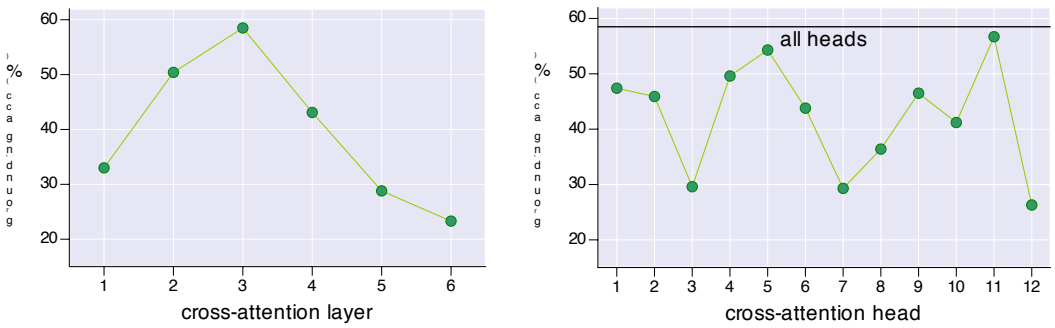
Figure 8: Grounding accuracy on the validation set of RefCOCO+. (a) varying cross-attention layers where each layer uses all heads. (b) varying cross-attention heads in the best-performing (3rd) layer.



Figure 9: Grad-CAM visualization on the cross-attention maps corresponding to individual words.

B Additional Per-word Visualizations

In Figure 9, we show more visualizations of per-word Grad-CAM to demonstrate the ability of our model to perform visual grounding of objects, actions, attributes, and relationships.



(a) (b)

图 8：RefCOCO+ 验证集上的定位准确率。(a) 改变交叉注意力层数，其中每层使用所有注意力头。(b) 在表现最佳的第 3 层中改变交叉注意力头数。



图 9：与单个词对应的交叉注意力图上的 Grad-CAM 可视化。

B 额外的逐词可视化

在图 9 中，我们展示了更多逐词 Grad-CAM 的可视化结果，以证明我们的模型对对象、动作、属性和关系进行视觉定位的能力。

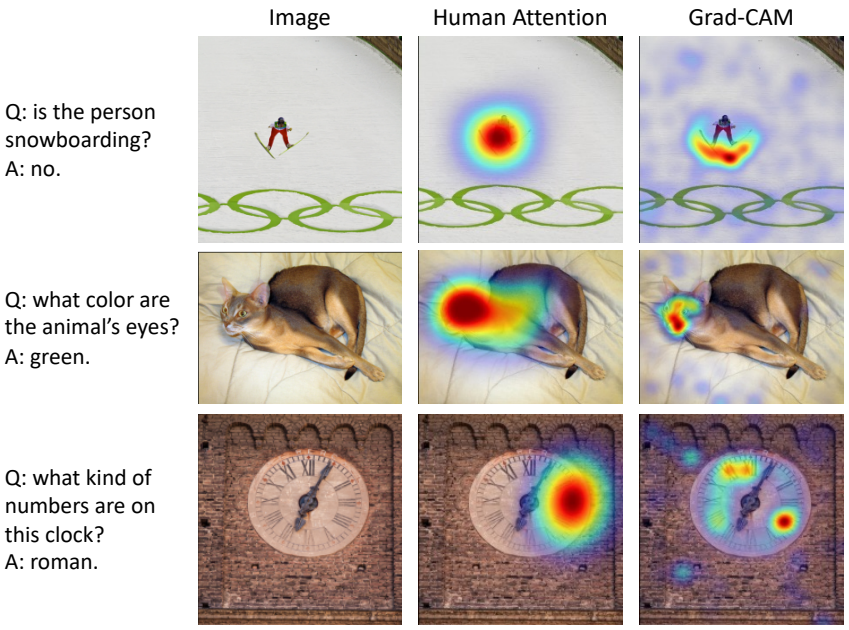


Figure 10: Qualitative comparison between human attention and ALBEF's Grad-CAM for VQA.

C Comparison with Human Attention

Das *et al*. [63] collected human attention maps for a subset of the VQA dataset [20]. Given a question and a blurred version of the image, humans on Amazon Mechanical Turk were asked to interactively deblur image regions until they could confidently answer the question. In this work we compare human attention maps to Grad-CAM visualizations for the ALBEF VQA model computed at the 3rd multi-modal cross-attention layer on 1374 validation question-image pairs using the rank correlation evaluation protocol as in [63]. We find Grad-CAM and human attention maps computed for the ground-truth answer to have a high correlation of 0.205. This shows that despite not being trained on grounded image-text pairs, ALBEF looks at appropriate regions when making decisions. Qualitative examples showing the comparison with human attention maps can be found in Figure 10.

D Additional Examples of Pseudo-targets

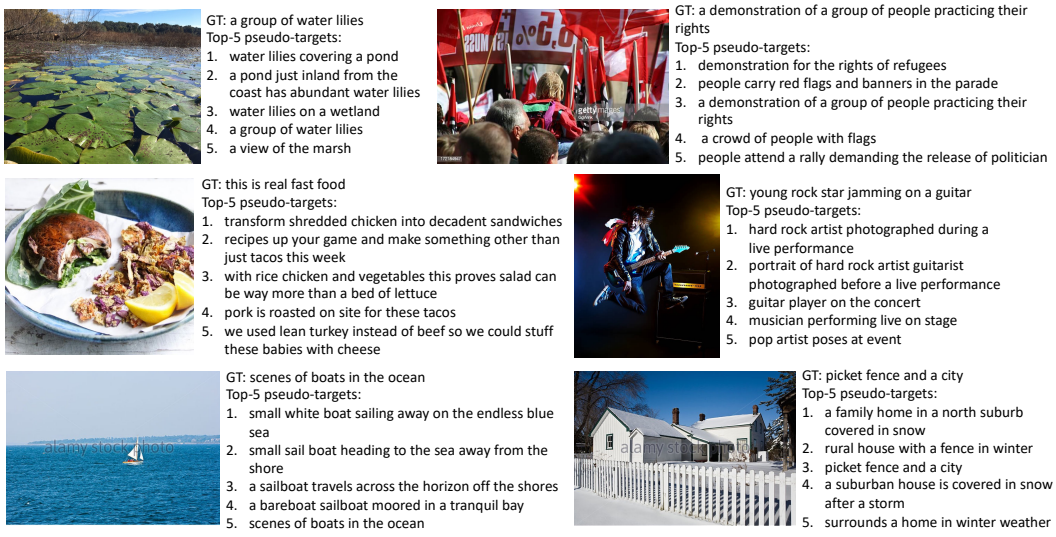


Figure 11: Examples of the top-5 most similar texts selected by the momentum model for ITC.

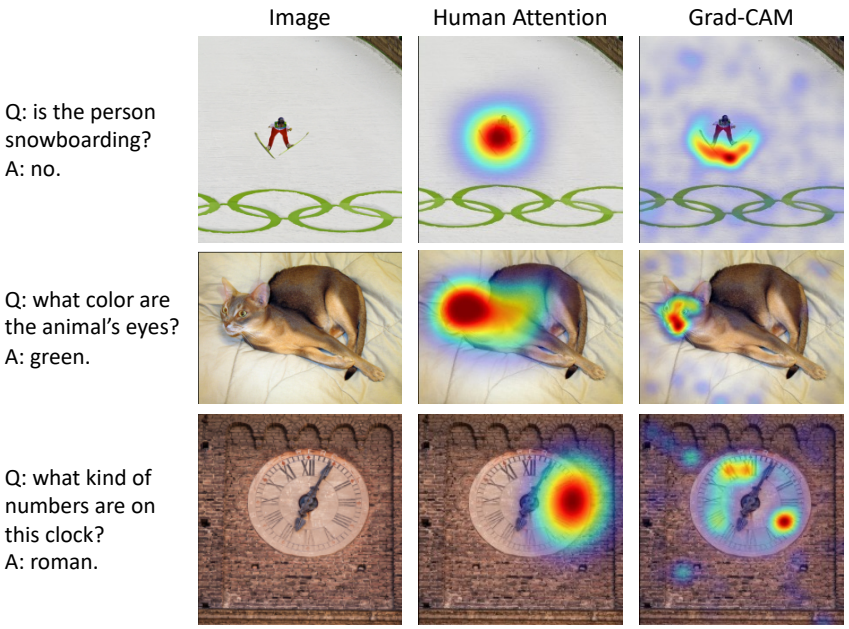


图 10：VQA 中人类注意力与 ALBEF 的 Grad-CAM 的定性比较。

C 与人类注意力的比较

Das *et al*。[63] 为 VQA 数据集 [20] 的一个子集收集了人类注意力图。给定一个问题和图像的模糊版本，Amazon Mechanical Turk 上的人类参与者被要求以交互方式对图像区域进行去模糊，直到他们能够有把握地回答该问题。在这项工作中，我们使用与 [63] 相同的秩相关评估协议，在 1374 个验证集问题-图像对上，将人类注意力图与在第 3 个多模态交叉注意力层计算得到的 ALBEF VQA 模型的 Grad-CAM 可视化结果进行比较。我们发现，针对真实答案计算的 Grad-CAM 与人类注意力图具有较高的相关性，相关系数为 0.205。这表明，尽管 ALBEF 没有在有定位标注的图像-文本对上进行训练，但它在做出决策时会关注适当的区域。展示与人类注意力图比较的定性示例见图 10。

D 伪目标的其他示例

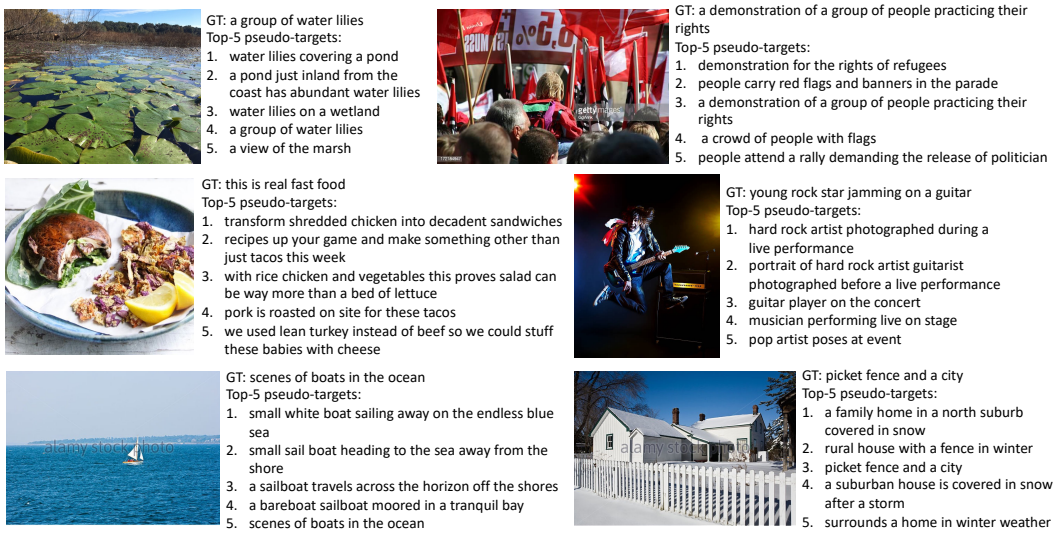


图 11：动量模型为 ITC 选择的前 5 个最相似文本示例。



Figure 12: Examples of the top-5 words generated by the momentum model for MLM.

E Pre-training Dataset Details

Table 8 shows the statistics of the image and text of the pre-training datasets.

	COCO (Karpathy-train)	VG	CC	SBU	CC12M
# image	113K	100K	2.95M	860K	10.06M
# text	567K	769K	2.95M	860K	10.06M

Table 8: Statistics of the pre-training datasets.

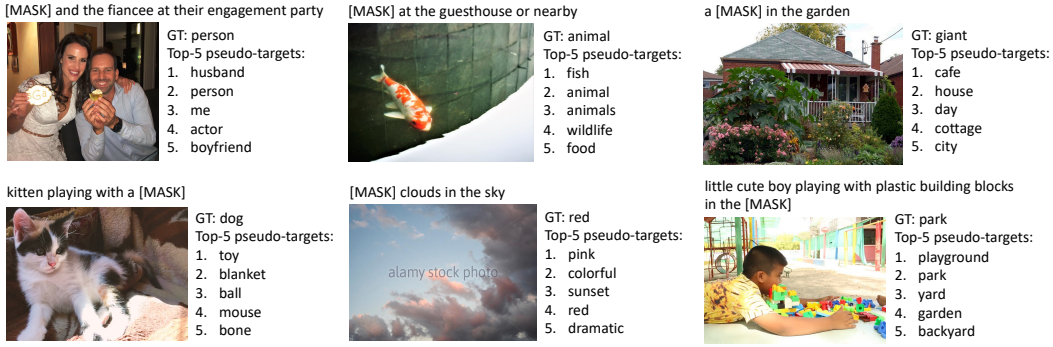


图 12：动量模型为 MLM 生成的前 5 个词示例。

E 预训练数据集详情

表 8 显示了预训练数据集中图像和文本的统计信息。

	COCO (Karpathy-train)	VG	CC	SBU	CC12M
# image	113K	100K	2.95M	860K	10.06M
# text	567K	769K	2.95M	860K	10.06M

表 8：预训练数据集的统计信息。